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Supervisor:

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Candidate Graceful Mulenga
Student ID: 23019111
Thesis Titled: Classification Models for Breast Cancer Brain Metastasis: An Integration of Neuroimaging and Genomics Data for Better Clinical Practice

I **Graceful Mulenga**, certify that the Thesis is all my own original work and that I have not obtained a degree in this University or elsewhere on the basis of any of this work.

Signed: ------------------------------ Date: 30 April 2026

Primary Supervisor

I **Chipo Zidana**, hereby confirm that I have inspected the above titled thesis and, to the best of my knowledge, it is based on the original work of the candidate.

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Abstract

Breast cancer brain metastasis (BCBM) is among the most aggressive forms of advanced breast cancer and is associated with poor prognosis. Although BCBM diagnosis can be through conventional approaches including invasive biopsies and standard imaging techniques through classification problem, these methods often pose a number of challenges including failure to capture the biological and structural complexity of metastatic tumors. Furthermore, when neuroimaging features are used as predictors to detect BCBM tumor from primary tumor in classification models , classical statistical methods such as logistic regression often fail due to the curse of dimensionality, where the number of features exceeds the number of observations ($p > n$).

Objectives and Method: This study aimed to address these problems associated with high-dimensional through a dual feature selection methods namely: 1) The Least Absolute Shrinkage and Selection Operator (LASSO) regularisation to enforce model parsimony by penalising and shrinking irrelevant coefficients in a logistic regression model, 2) the second method is a biologically informed strategy, where the fusion of genomic data (GSE125989 from the Gene Expression Omnibus (GEO)) with neuroimaging data (BCBM-RadioGenomics (p=107) from The Cancer Imaging Archive (TCIA)) to inform feature selection. Differentially expressed genes between primary tumors and BCBM were identified using Linear Models for Microarray Analysis (LIMMA). These genes were then used as transcriptomic signatures to assess correlations with brain regions associated with BCBM using the UK Biobank. Specifically, only neuroimaging features from these identified were then selected as biologically relevant features for BCBM classification. Additionally, the study explored different machine learning classifiers that helped address problems of high dimensionality including multicollinearity, overfitting and sparsity among others. These machine learning models namely, support vector machine (SVM), random forest, and eXtreme Gradient Boosting (XGBoost) were trained across three data sets (i) Original BCBM radiogenomics data (baseline models), (ii) LASSO reduced data were features were selected using LASSO regression , and iii) genom-identified models that used only features biologically informed by genes.

Results: Differential expression analysis identified 105 significant probesets corresponding to 79 unique genes differentially expressed between primary breast cancer and BCBM. Out of these 79 genes, only 17 were available in the UK Biobank and were then used to inform the gene feature selection. Across all the models (baseline, LASSO-based, genom-informed), XGBoost and random forest classifiers were top performers achieving an AUC of 84% and 83%, respectively for the baseline model compared to 79% and 75% for SVM and LASSO logistic regression. Additionally, the results showed that the introduction of LASSO regularisation and gene-informed selection methods reduced the feature space by 77% and 69%, respectively. This reduction in the feature space affected model performance differently, with AUC improvements of 5%, 3%, 7%, and 10% for XGBoost, random forest, SVM, and LASSO logistic regression, respectively, following LASSO-based feature selection compared to the baseline models. Although the genom-identified selection criteria had marginal lower performance compared to baseline models, per-

formance across all classifiers and metric were almost the same. Moreover, the genom-identified method major predictors were such that they correlate with specific genomic markers. Our model identified intensity and texture features major predictors using genom-idenfied selection method whilst LASSO based was dominated by shape-based features. It was concluded that although mathematical method via LASSO regularization produced higher performance across classifiers, radiogenomic improved model stability and biological interpretability acoss all the models. Additionally, radiogenomic approach reduced the black-box nature of machine learning models by providing biologically and explainable features which may support the development of more targeted and clinically relevant treatment strategies.

Dedications

I dedicate this thesis to my late mother who remains my inspiration.

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1. Introduction

1.1 Background

Breast cancer is one of the most commonly diagnosed malignancies among women and remains one of the leading causes of cancer-related mortality [Giannoudis et al., 2024]. It comprises multiple molecular subtypes, including luminal A, luminal B, human epidermal growth factor receptor 2 (HER2)-positive, and triple-negative breast cancer (TNBC). Each subtype is associated with different prognoses, treatment responses, and metastatic behavior [Kallah-Dagadu et al., 2025, Zhang et al., 2025]. This heterogeneity presents significant challenges in the classification, prediction, diagnosis, and clinical management of breast cancer, particularly in advanced stages. One of the most serious complications of breast cancer is its potential to metastatically spread to the brain. Metastasis is the spread of cancer cells from the place where they first formed to another part of the body [Zwanenburg et al., 2020a]. Although cancer can spread to various parts of the body, breast cancer commonly metastases to the brain, liver, lungs, and bones [Boire et al., 2020]. The brain is a very sensitive organ, such that even small metastatic lesions can lead to cognitive and functional complications [Garrone et al., 2024]. Brain metastases happens when cancer cells from the primary breast tumor spread to the brain, usually through the bloodstream [Boire et al., 2020]. This is especially common in aggressive types like HER2-positive and triple-negative breast cancer because they are more likely to cross the blood-brain barrier (BBB) and grow in brain tissue [Rostami et al., 2016].

Genomic data, especially gene expression profiles, helps us understand how breast cancer spreads to the brain (BCBM). According to National Human Genome Research Institute [2023], genomic profiling shows how certain genes (segments of DNA that carry instructions for making proteins) allow tumor cells to adapt to the brain and form new colonies. For example, when genes like HER2, cyclooxygenase-2 (COX2), heparin-binding EGF-like growth factor (HBEGF), and ST6GALNAC5 are overexpressed, breast cancer cells are more likely to cross the blood–brain barrier and create new tumors in the brain [Bos et al., 2009, Fan et al., 2020, Zhang et al., 2025]. These genes are also responsible for cell adhesion, immune evasion, and forming new blood vessel. Gene expression data can also show which brain regions are more likely to be affected by metastasis. Some genes related to nerve connections, brain signaling, and blood vessel changes are linked to specific parts of the brain. This may help explain symptoms like movement, memory, or sensory problems, depending on where the tumor spreads [Boire et al., 2020]. Moreover, this information can help predict how the tumor will behave and what neurological effects BCBM may cause. However, genomic data alone do not show exactly where the tumor is in the brain and usually require a biopsy, which is invasive [Fan et al., 2020]. It can also miss differences between tumor areas, how cancer cells interact with brain tissue, and the challenge of connecting gene expression to brain function problems [Leone et al., 2015]. Therefore, imaging, especially magnetic resonance imaging (MRI), is still essential and is considered the gold standard for visualizing metastatic lesions and providing spatial context into genomic data. Traditional methods such as MRI are important for diagnosing and monitoring BCBM. This is done through neuroimaging, which is extracting quantitative features from MRI to capture tumor shape, texture, intensity, and many other. However, these methods have limitations in detecting metastases or showing tumor differences. While neuroimaging features often lack direct biological meaning, they do provide spatial information that helps explain tumor biology [Gillies et al., 2016]. They can also help identify physical features of BCBM, but they provide limited information into neurological functions and the structural effects of these metastases on the brain.

The diagnosis of BCBM from neuroimaging data can be done using a classification problem where neurological features are used to distinguish the primary breast tumor from the BCBM tumor through classical statistical methods like logistic regression. However, neuroimaging data is often regarded as high dimensional data where the number of variables exceeds the number of observations ($p > n$). High dimensionality (curse of dimensionality) often poses challenges including data sparsity , multicollinearity, overfitting to traditional classification models due change in the structure of the covariance matrix. It is therefore imperative to explore robust methodologies that improve BCBM tumor detection. This can be achieved either through biologically or mathematically driven feature selection techniques to address high dimensionality, or through the application of advanced classification methods capable of handling non-linearity, sparsity, and overfitting.

This work seeks to explore a dual approach of dimension reduction strategy that addresses these challenges. The first approach involves employing a regularisation on the logistic regression classifier through a Least Absolute Shrinkage and Selection Operator (LASSO). This method enforces model parsimony by penalising irrelevant coefficients hence shrinking the number of features needed to determine a classification outcome. Although, the LASSO can be a helpful parsimonious model, the biological interpretation of the selected features may not be biologically interpretable. Therefore, this research introduces a second option that uses the fusion of genomic and neuroimaging data through the implementation of latent statistical correlations between genetic signatures and neuroimaging phenotypes. This process is often referred to as radiogenomics. These identified correlations are then used to identify anatomically relevant brain regions (Regions of Interest (ROIs)) mainly affected by BCBM. As alluded by [Gallivanone, 2022], radiogenomics provide an effective and non-invasive way to classify tumors whilst dealing with high dimensionality and interpretability issues hence tumor detection maybe improved. Furthermore, by combining genomics with imaging, radiogenomics helps connect visible imaging features to the biological processes that drive tumor growth and spread [Fan et al., 2020].

Machine learning (ML) classification models, including support vector machines, LASSO-based models, and ensemble techniques, can efficiently handle high-dimensional

data. These methods mitigate overfitting through regularisation and effectively manage sparsity, providing a robust framework for analysis of high-dimensional data. These approaches can improve model stability and enable effective variable selection, thereby enhancing classification accuracy [Chen and Guestrin, 2016, Khyathi et al., 2025, Tibshirani, 1996]. Furthermore, ML models are useful for recognising patterns, identifying relevant features, and performing classification tasks [Erickson et al., 2017]. Combining methods that employs both an effective and biologically interpretable feature selection in high dimensional data and robust classification through the use of ML may represent an important step in BCBM diagnosis.

The work aims to use the power of radiogenomics as a biologically interpretable feature selection method together with a purely mathematical selection of LASSO method as a dimension reduction strategy. It seeks to explore which method yields a more stable and generalizable decision manifold for machine learning classifiers, ultimately enhancing the diagnostic accuracy and interpretability of BCBM detection.

Problem statement

BCBM is associated with poor survival outcomes, hence the need for improved diagnosis and treatment. Early BCBM tumor detection from primary breast cancer tumor is normally done through a classification model using neuroimaging features. This classification has often proved to be either difficult or inaccurate due the curse of dimensionality associated with neuroimaging data leading to either unstable decision boundaries and overfitting. Secondly, it is also possible that many features than need maybe associated with BCBM tumor making it difficult to make meaningful physiological interpretations of the results. It is therefore useful to implement variable selection methods that does not only reduce dimension of data but robust to improve both model accuracy and interpretability. Whilst mathematical models like LASSO may provide a sparse linear solution, it may however fail to capture the complex, non-linear interactions between neuroimaging data (MRI-T2-weighted) texture features and genetic expressions. Radiogenomics on the other hand provide a promising way that deal with both dimension reduction and biological interpretability. Therefore, there is a need to evaluate whether a radiogenomics machine learning classification pipeline which constrains feature selection on identified ROIs from the biological correlations yields a more effective, accurate and statistically separable decision manifold compared to traditional classification methods like LASSO logistic regression.

Aim

To develop and evaluate high-dimensional Machine learning classification frameworks for BCBM tumor detection through the comparison of different feature selection methods.

Specific objectives

1. To develop baseline classification models for BCBM tumor detection using a variety of machine learning methods (RF, XGBoost, SVM).
2. To implement regularised feature selection on BCBM MRI image data using LASSO-based shrinkage and analyze its effects on tumor classification.
3. To identify differentially expressed genes (DEGs) associated with metastatic progression, utilizing multiple testing corrections to ensure the features used for classification are statistically significant.
4. To develop and implement BCBM classification framework that utilises only neuroimaging features that are from most discriminative brain Regions of Interest (ROIs) identified through associations between DEGs T2-weighted MRI features through multivariate correlation analysis.
5. To perform a comparative classification assessment for BCBM tumor detection between gene-informed classification models against the LASSO-selected models.
6. To identify interpret the biological and physiological relevance of selected features by linking them to brain function.

Research Questions

1. Which non-linear machine learning classifier (SVM with RBF kernel or Tree-based ensembles) significantly improve the classification accuracy compared to frequently used logistic regression with LASSO?
2. Does the use of only LASSO selected features in a non linear classifier a improves distinct statistical separation between metastatic and non-metastatic tumor classes and hence diagnosis?
3. What are the major genes associated with BCBM and how we incorporate knowledge of the gene function into feature selection criteria for BCBM tumor classifier?
4. Does integrated gene-neuro-imaging feature selection criteria improve classification performance compared to baseline models in objective 1.
5. How does the generalization error differ when using a mathematically sparse feature set (LASSO) versus a biologically informed one (Radio-genomics)?
6. What is the biological and physiological relevance of the selected features in relation to brain function in BCBM?

7. **To what extent do SHAP feature attribution values reveal consistent diagnostic markers across the different classification architectures?**

Significance of the study

The significance of this study lies in its objective to develop a non-invasive, machine learning-based radiogenomic classification system for the early detection and classification of brain metastases of breast cancer (BCBM). The brain is a critical organ responsible for essential functions such as cognition, motor control, sensation, and memory, and even small metastatic lesions can severely disrupt these functions. Traditional diagnostic methods often fail to detect early metastatic changes or capture the full biological complexity of tumors. By integrating genomic data with neuroimaging features, this study offers a novel approach to identify features that reflect the molecular and spatial characteristics of metastatic tumors. **A methodological contribution of this study is the use of variable selection approaches based on both machine learning (LASSO logistic regression) and gene-identified neuroimaging feature selection. This approach supports improved classification performance by selecting neuroimaging features that reflect gene expression patterns, while also enhancing biological relevance and clinical interpretability. This contribution relies on the application of robust statistical methods to ensure that feature selection is both reliable and reproducible. Furthermore, the study addresses the need for biologically guided feature selection within high dimensional datasets and the development of classification models that provide biologically meaningful information to affected brain regions. As a result, this not only improves diagnostic accuracy, but also provides information on tumor behavior, metastatic potential, and its neurological impact on brain function and structure. Overall, the findings contribute to the development of clinically relevant, robust models that support precision medicine in cancer studies and can inform personalized treatment strategies.**

2. Literature review

Breast cancer is a leading cause of cancer-related illness among women globally, with a notably high burden reported in Sub-Saharan Africa [Giannoudis et al., 2024]. The disease is classified into several molecular subtypes namely luminal A, luminal B, HER2-positive, and triple-negative breast cancer (TNBC) each of which is identified through immunohistochemical testing for estrogen receptor (ER), progesterone receptor (PR), and HER2 expression [Cheng and Hung, 2007]. A major clinical concern across all subtypes is the tendency of breast cancer cells to spread to distant organs(metastases) including the liver, bone, lungs, and brain, each presenting its own management challenges [Cha, 2009, McCormack et al., 2020].

2.1 Breast Cancer Brain Metastasis

Brain metastasis occurs as the initial site of distant spread in approximately 12% of breast cancer patients, making breast cancer the second most common source of brain metastases overall [Orrantia-Borunda et al., 2022]. Among patients with advanced disease, the reported risk ranges from 10 to 16% in living cohorts and rises to around 30% in autopsy studies [Lundberg and Lee, 2017]. Outcomes in this population are generally poor, largely because the blood-brain barrier (BBB) restricts the entry of most systemic therapies into brain tissue, limiting treatment effectiveness [Boire et al., 2020]. Brain metastases tend to occur more frequently in younger patients and in those with larger tumors, higher nuclear grade, or specific subtypes such as oestrogen receptor negative, triple-negative, and HER2-overexpressing breast cancer, as well as in patients with nodal involvement [Boire et al., 2020]. Standard imaging tools used to detect and characterise these lesions include magnetic resonance imaging (MRI), computed tomography (CT), and positron emission tomography (PET), each providing information on tumor size, number, and location [Demetriou et al., 2024, Medicine, 2025].

Alongside imaging, a neurological examination is often done to assess the patient's symptoms and brain function. This exam helps doctors identify any weakness, memory problems, or other changes caused by the tumor. Over the years, outcomes for patients with BCBM have improved and are now generally better than for patients with brain metastases from lung cancer [Dubey et al., 2023]. However, BCBM is serious medical challenge and can lead to mortality. Moreover, BCBM often causes significant changes in brain function, depending on the tumor's size and location, therefore, it is crucial to investigate its effects on the nervous system. These changes directly influence the quality of life and treatment choices. It is also crucial to learn more about how tumor biology, including genetic changes, relates to the physical and functional effects of BCBM. This knowledge can help improve early detection and guide more personalized treatments.

2.1.1 Neurological Impacts From BCBM

BCBM leads to poor survival and affects quality of life by causing neurological problems. Patients with brain metastases often experience symptoms such as headaches, memory or thinking problems, weakness, numbness, seizures, and behavioral changes [Boire et al., 2020]. For example, problems with thinking and mood are commonly seen when the tumor affects the frontal lobe, while issues with movement and coordination are more frequent when the parietal lobe or cerebellum is involved [Gallivanone, 2022, Niwińska et al., 2010]. Seizures are more likely when the tumor is located in the outer parts of the brain (the cortex), especially if the lesion has bleeding or swelling around it [Nayak et al., 2012]. Larger tumors can press on nearby brain tissue and cause swelling (edema), which increases pressure in the skull. Because of these serious effects, BCBM can greatly lower a patient's quality of life and make treatment more difficult. This shows why early detection, accurate diagnosis, and personalised treatment are very important in the care of patients with BCBM. Although neurological impacts show the clinical effects of BCBM, there is still need to explore the genomic alterations that contribute to tumor spread in the brain.

2.2 Genomics

Genes are functional units of DNA that carry instructions governing cell behaviour and biological traits, and alterations in these genes, along with changes in associated proteins and cellular pathways, play a central role in driving cancer growth and metastatic spread [Pearson, 2006]. These genomic changes are particularly relevant to the development and progression of BCBM. Among the recognised breast cancer subtypes, HER2-positive and TNBC carry a notably elevated risk of brain involvement, due to their aggressive biological behaviour and distinct patterns of genomic alteration [Clough and Barrett, 2016, Medicine, 2025]. These alterations equip tumor cells with the capacity to survive within and colonise the unique microenvironment of the brain [Lin and Winer, 2007]. Several molecular pathways have been implicated in promoting brain tropism which is the preferential ability of cancer cells to invade and replicate in the brain. For example, in TNBC, high expression of genes involved in the epithelial-mesenchymal transition (EMT), cell adhesion, and chemokine signaling has been linked to brain metastasis [Olson and et al., 2013]. Understanding the biological mechanisms behind cancer needs the analysis of the genomic alterations that facilitate metastatic spread and adaptation to the brain. This is where genomic profiling (the identification of key mutations and expression patterns that drive tumor progression), is essential in the study of breast cancer. These mutations are identified from deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) sequences extracted from tumor tissue samples [Rossing et al., 2019]. HER2 amplification is one of the most common genetic changes assoocioated to the spread of breast cancer to the brain. Targeted treatments like trastuzumab, lapatinib, and T-DM1 have been developed for HER2-positive cancers, however, many of these drugs do not enter the brain effectively, which allows HER2-positive cancer cells to survive or return in the brain. Gene expression studies have also identified other genes that may help breast cancer cells reach the brain. These include COX2 (also called PTGS2), the EGFR ligand HBEGF, and the enzyme ST6GALNAC5

[Bos et al., 2009]. These genes help cancer cells pass through the blood-brain barrier. For example, breast cancer cells that produce more ST6GALNAC5 are better able to stick to the brain's blood vessels and move into brain tissue. This behaviour is usually not seen in breast cancer cells that spread to other parts of the body. These findings have helped researchers better understand the different types of genetic changes that occur in BCBM. They also offer new ways to improve personalised treatment, such as predicting patient outcomes, selecting treatment groups, and tracking how patients respond over time. For example, HER2 mutations have been discovered in metastatic brain tumors, which may open up new treatment options. In addition it was found that less invasive methods like liquid biopsies, using blood samples to analyse tumor DNA or circulating tumor cells, have been used to detect the presence of BCBM and monitor resistance to treatment early and noninvasively [Curtis, 2015]. However, genomic profiling still faces several limitations which include high costs, limited access to tumor tissue, and heterogeneity of the tumor [Gerlinger et al., 2012]. tumor heterogeneity is the coexistence of genetically, molecularly, or cellularly distinct populations within the same tumor, or between the primary tumor and its metastatic sites [Carvalho et al., 2025, Gerlinger et al., 2012]. This diversity can affect how cancer grows, spreads, and responds to treatment. The heterogeneity between primary and metastatic sites makes interpretation challenging, and obtaining tissue from the brain is often invasive and impractical. Moreover, genomic data alone cannot provide information on the physical characteristics of brain metastases, such as their size, number, or exact location in the brain as mentioned above. These features are important for guiding diagnosis and treatment planning [Curtis, 2015]. Because genomics alone cannot capture structural tumor features, neuroimaging particularly MRI, plays a supplementary role. Additionally, It remains the gold standard for detecting and monitoring brain metastases and collectively, these approaches form the foundation for radiogenomics.

2.3 Magnetic Resonance Imaging and BCBM Diagnosis

MRI is the imaging technique preferred for detecting brain metastases due to its ability to provide both structural and functional information [Gallivanone, 2022]. Common MRI findings in brain metastases include multiple, clearly defined, contrast-enhancing lesions with surrounding swelling (vasogenic edema). There is advance methods used in MRI, such as dynamic contrast-enhanced MRI (DCE) and diffusion-weighted imaging (DWI), which offer additional information into tumor blood flow and cell density. MRI also detects cancer that may not be found using physical exams, mammograms, or ultrasound [Yeh et al., 2019]. neuroimaging extends MRI by systematically deriving large sets of quantitative descriptors from imaging scans to describe the tumor's shape, size, internal structure, and enhancement behaviour. These include morphological features such as roundness, irregularity, smooth and spiculated edges and intensity-based features such as mean, variance, and entropy [Perou et al., 2000]. Texture features are especially important for understanding tumor heterogeneity. These are calculated using several families of feature extraction methods, including Gray-Level Co-Occurrence Matrix (GLCM), captures different pixel intensities, Level Run Length Matrix (GLRLM), measures the length of consecutive pixels with the same intensity, Gray-Level Size Zone Matrix (GLSZM), quantifies the size of connected regions with the same intensity, Neighbourhood Gray-Tone Difference Matrix (NGTDM), which captures the difference between a pixel and its neighbours and Gray-Level Dependence Matrix (GLDM), assesses the gray level's dependence in a neighbourhood [Zwanenburg et al., 2020b]. These texture features are used to quantify tumor heterogeneity and can help show important biological behaviour. Features that are dynamic, such as those from DCE-MRI (wash-in/wash-out rates, time-to-peak, and kinetic curve patterns), describe tumor vascularity and perfusion. These features can be extracted using software tools in R or Python [Erickson et al., 2017]. A meta-analysis of 19 studies, that enrolled 2,610 breast cancer patients found that MRI identified additional disease in 16% of cases (IQR 11–24). Another large study by Morrow et al. [2011] reported that 3,252 women showed that MRI could detect cancer in the opposite breast that other methods had missed. Furthermore, some MRI patterns have been linked to genetic changes. For example, tumors that show uneven or heterogeneous enhancement on DCE-MRI have been associated with high-risk genomic profiles related to tumor growth and angiogenesis [Saha et al., 2018, Yeh et al., 2019]. These findings suggest that neuroimaging features from MRI could act as non-invasive biomarkers (observable features that reflect tumor characteristics).

However, MRI also has its limitations. Firstly, it does not provide information that can identify the genetic causes or biological behaviour of metastatic tumors. Secondly, some lesions can be missed or misinterpreted, especially in early stages or when they are in hard-to-reach areas of the brain. This can therefore cause false negatives or false positives [Gillies et al., 2016]. For example, small metastases in the leptomeninges or micrometastatic lesions may not be visible even with contrast [Park et al., 2023]. Additionally, Cha [2009] observed that changes caused by treatment, like radiation damage or pseudoprogression, can look similar to active tumor tissue, making it difficult to interpret MRI results. Because of these limits, researchers are now combining imaging with genomic data. This combined approach, also known as radiogenomics, offers a more complete way to study brain metastases and guide treatment.

2.4 Radiogenomics

Radiogenomics is a growing field that links the physical characteristics observed in medical images referred to as the neuroimaging phenotype or radiophenotype, with the underlying genomic profile of a disease [Gallivanone, 2022]. The genomic profile encompasses gene expression patterns, mutations, and other genome-related features, with the broader aim of improving tumor classification and biological understanding [Gallivanone, 2022]. The field integrates neuroimaging features with genomics, placing particular emphasis on gene expression profiling, which identifies which genes are active within a tumor and can help distinguish tumor types as well as estimate the likelihood of metastatic spread to the brain [Barbadilla-Martínez et al., 2025]. A central motivation for combining these two data sources is that imaging captures spatial and structural information about the entire tumor volume that genomic data alone cannot provide, since gene expression measurements are derived from small tissue samples or averaged across multiple tumor regions and may therefore fail to reflect the full extent of intratumoral heterogeneity [Mazurowski, 2015]. Imaging, by contrast, captures this variability across the whole tumor, offering complementary biological knowledge that strengthens both classification accuracy and clinical interpretability when used alongside genomic information [He et al., 2024, Pearson, 2006]. This method has already shown successful prognostic contribution. Integrating MRI-derived features with genomic data has been shown by recent work to be successful as it was able to improve survival prediction in glioblastoma and other primary brain tumors [Cha, 2009]. Another example was by Yan et al. [2011], who reported that, a radiomics MRI-model successfully identified patients with EGFR amplification, MGMT promoter methylation, and certain glioblastoma subtypes, with a high performance in predicting overall survival. Moreover, it has been shown that combining MRI features with genomic information can help predict dysregulation in key oncogenic pathways, which in turn contributed to precision medicine strategies for managing glioblastoma [Sanchez and Rahman, 2024]. In primary brain tumors and BCBM, radiogenomic analysis has also associated certain neuroimaging patterns to mutations in genes such as VHL, PBRM1, SETD2, KDM5C, and BAP1 - genes known to be associated with higher tumor grades, advanced disease stage, and reduced survival [Karlo et al., 2014]. Similar findings have been reported in breast and lung cancer studies, where neuroimaging features showed correlations with hormone receptor profiles and major oncogenic drivers [Fan et al., 2020, Garrone et al., 2024]. Furthermore, through these findings, the understanding of tumor biology at gene level has been deepened and also provided imaging markers that assist with treatment practices and prognosis prediction [Zhu et al., 2019]. They have also shown the importance of radiomics in identifying neuroimaging features that can be detected through the human eye.

However, radiogenomics studies in brain metastasis particulalry BCBM remain limited. BCBM is challenging because of where it is located, its neurological effects and generally poor clinical outcomes. Establishing associations between neuroimaging features and gene expression could provide clarity on how metastatic lesions disturb specific brain regions and eventually lead to neurological symptoms [Fan et al., 2020]. There is a development in this direction which is the introduction of the `NeuroimaGene` R package Taha et al. [2025]. This package systematically maps DEG's onto brain regions characterized through neuroimaging. Neuroimaging features affected by genetically regulated expression (GReX) of selected genes or gene sets are easily identified by `NeuroimaGene`. It does so by integrating predicted gene activity with neuroimaging data to generate genome-informed neuroimaging features [Bledsoe and Gamazon, 2024]. For example, over 3500 brain imaging features were analyzed by Yeh et al. [2019], showing that the gene ICA1L was associated with variations in the superior frontal gyrus and caudate regions which are both involved in cognitive and motor processes. This study illustrated how radiogenomics is able to uncover certain genes may influence brain structure, aiding in the explanation of broader neurological and biological disorders. Despite these promising knowledge, a number of challenges continue to limit radiogenomics. One difficulty is correlating gene activity with specific neuroimaging features because of high dimensional data [Mazurowski, 2015]. An effective integration of large genomic and imaging datsets requires advanced statistical and computational methods [Incoronato et al., 2017, Wang et al., 2025]. Even though current techniques can identify associations, a lot of findings are correlational not causal, which makes biological and clinical interpretation difficult. Therefore the combination of radiogenomic data with advanced models allows the development of more accurate predictive models that can assist in the classification of BCBM.

2.5 Classification Modelling in BCBM

Combining neuroimaging features with genomic data has been explored using statistical and machine learning models to classify BCBM [Fan et al., 2020, Luo et al., 2022, Vapnik, 1964]. However, radiogenomic classification presents several statistical challenges, particularly because of the high dimensionality of the data [Aerts et al., 2014]. Neuroimaging feature extraction can produce hundreds or thousands of imaging variables describing tumor intensity, shape, texture, and spatial heterogeneity, while genomic datasets may contain thousands of gene expression measurements [Edgar et al., 2002a]. As a result, radiogenomic analysis often fall into a high-dimensional where the number of predictors exceeds the number of observations ($p > n$). Under these conditions, classical statistical estimation methods can become unstable and produce unreliable parameter estimates [Harrell et al., 2001, Ruppert, 2004]. For example, when maximum likelihood estimation is applied in such, coefficient estimates may have large variance and models may overfit the training data, leading to poor classification performance in independent datasets [Babyak, 2004]. Another challenge arises from multicollinearity among predictors. Many Neuroimaging features capture related aspects of tumor heterogeneity, including first-order intensity measures, grey-level co-occurrence matrices, run-length matrices, and shape descriptors [Orton et al., 2023, Zeng et al., 2022]. Because these variables are derived from similar image characteristics, they are often strongly correlated [Gillies et al., 2016, Zwanenburg et al., 2020a]. High correlation among predictors can increase the variance of parameter estimates, reduce model stability, and make it difficult to interpret the contribution of individual features [Orton et al., 2023]. Additionally, limited sample sizes present an additional difficulty. Studies investigating BCBM involve relatively small cohorts because datasets that include both imaging and genomic information are rare. Small sample sizes increase the likelihood of overfitting, particularly when complex models with many parameters are fitted [Babyak, 2004]. Moreover, class imbalance may occur when primary breast cancer and BCBM cases are not equally represented, which can bias classification models towards the majority class. Together, these challenges emphasise the need for statistical methods that can handle high-dimensional feature spaces, correlated neuroimaging features, and limited sample sizes. Methods such as penalized regression models, support vector machines, and ensemble tree-based algorithms have been widely adopted in radiogenomic studies for this reason [Erickson et al., 2017, Karlo et al., 2014, Park et al., 2019]. Combining methods tha capture non linearity, handle overfitting and deals with the curse of dimensionality in classifying high dimension data like neuroumaging data can be very helpful.

2.5.1 Classification with LASSO Logistic Regression

Logistic regression is a commonly used statistical method for analysing binary outcomes in biomedical research. The goal is to model the probability of an event occurring as a function of one or more predictor variables [Tibshirani, 1996]. The model achieves this through the logit link function, which transforms the probability of the outcome into log-odds and expresses it as a linear combination of the predictor variables [Hosmer Jr et al., 2013]. An advantage of logistic regression in clinical research is that its estimated coefficients can be directly related to odds ratios, making the model outputs easy to communicate to both statistical and non-statistical audiences [Harrell et al., 2001]. This is particularly valuable in medical settings where clinical interpretability is essential for informing treatment decisions [Sloane and Morgan, 1996]. However, logistic regression becomes problematic in high-dimensional radiogenomic datasets. It relies on maximum likelihood estimation, which assumes that the number of observations is large relative to the number of predictors [Nelder and Wedderburn, 1972]. When $p \geq n$, this assumption is violated, and the Fisher information matrix may become singular, resulting in unstable parameter estimates [Hosmer Jr et al., 2013]. In such cases, coefficient estimates become unstable and the model may overfit the training data, leading to poor generalisation to new datasets [Sloane and Morgan, 1996].

Given the limitations of standard logistic regression in high-dimensional settings, regularisation-based approaches have been developed to address this challenge [Tibshirani, 1996]. One of the methods is the Least Absolute Shrinkage and Selection Operator (LASSO). LASSO introduces an L_1 penalty (the sum of the absolute values of the regression coefficients) to the regression objective function, which shrinks the absolute values of regression coefficients towards zero [Tibshirani, 1996]. This penalty encourages sparsity in the model by forcing some coefficients to become exactly zero. This removes non-informative predictors while retaining the most relevant variables. As a result, LASSO performs both classification and feature selection simultaneously, making it useful where predictors are a large number. In a study by Saha et al. [2018], LASSO was used to reduce 529 MRI features to 37, which signicicantly improved tumor status classification. Similarly, Li [2022] used LASSO to identify deep neuroimaging features extracteed denoising autoencoder, and showed that these features can predict tumor characteristics such as receptor status (ER, PR, HER2), lesion size, and lymph node involvement with AUC values exceeding 0.90 accurately. In another study, Zhu et al. [2019] and his colleagues showed that the deep neuroimaging features identified by LASSO showed stronger associations with genomic pathways and known primary breast cancer genes than traditional imaging features. This revealed the importance of identifying biologically relevant imaging features. LASSO has been further used beyond the tasks of classification [Li, 2022]. For example, a LASSO Cox regression has been used to predict survival with MRI neuroimaging features, revealing that reducing noise and irrelevant features can simplify models and further improve interpretability [Li, 2022]. However, even though LASSO has shown strong perfomance in breast cancer work, applications of it in BCBM specifically is limited. Past studies have focused on predicting patient prognosis rather than tumor status or even the combination of genomic and neuroimaging data Carvalho et al. [2019], Erickson et al. [2017], Tibshirani [1996].

2.5.2 Classification via Support Vector Machine

Support vector machines (SVM) represent another widely used approach for classification in high-dimensional datasets [Noble et al., 2004, Yu et al., 2023]. SVM, which was introduced by Cortes and Vapnik [1995], constructs a decision boundary that maximises the margin between two classes. The observations closest to the decision boundary, known as support vectors, determine the position of the separating hyperplane [Noble et al., 2004]. One of the advantages of SVM is the use of kernel functions, which allow nonlinear relationships between predictors and outcomes to be modelled without transforming the input variables [Schölkopf and Smola, 2002]. This is particularly important in radiogenomics, where the relationship between imaging features and tumor biology is unlikely to be purely linear. By mapping the data into higher-dimensional feature spaces, SVM can capture complex patterns in the data. Additionaly, SVMs have also been adopted in medical imaging because of their ability to handle small datasets with high-dimensional feature spaces. For example, Arefan [2020] used SVMs with 2D and 3D MRI features to predict distant metastasis in breast cancer, reporting AUC values of 0.80 and 0.76, along with high specificity and predictive value. These findings demonstrate that SVMs can effectively model imaging features for metastasis prediction. Moreover, Erickson et al. [2017] showed that SVM has been widely used in radiology applications for tumor detection, classification, and prognosis prediction due to its ability to capture complex nonlinear relationships. In breast cancer imaging, SVM models have been applied to MRI-based neuroimaging features to classify tumor subtypes and predict metastatic risk. It often achieves competitive performance compared with other machine learning algorithms [Saha et al., 2018].

Despite these advantages, SVM also has limitations that must be considered. Model performance is sensitive to the choice of kernel function and hyperparameters such as the cost parameter and kernel width [Schölkopf and Smola, 2002]. Tuning using cross-validation is therefore required to achieve optimal performance. In addition, SVM models are often less interpretable than regression-based approaches, which may limit their usefulness when clinical interpretation of model parameters is required [Ruppert, 2004]. However, incorporating variables selected through genomically informed approaches can improve both model performance and interpretability. Linking selected neuroimaging features to specific gene expression patterns and associated brain regions is beneficial in SVM. The resulting SVM model becomes more clinically meaningful, as the features can be related to biological functions and neurological processes in the brain [Gillies et al., 2016, Lo Gullo et al., 2020].

2.5.3 Random Forest

Random forest is an ensemble learning method introduced by Breiman [2001]. It constructs a large number of decision trees using bootstrap samples of the data and random subsets of predictors at each split [Angelone et al., 2024]. The final prediction is obtained by aggregating the predictions of individual trees, by using majority voting in classification tasks Park et al. [2019]. Random forest can capture nonlinear relationships and complex interactions among predictors without requiring explicit specification of these relationships. Secondly, it is relatively robust to noise and multicollinearity among predictors [Breiman, 2001, Ruppert, 2004]. In breast cancer imaging studies, random forest has been successfully applied to classify tumor subtypes and predict clinical outcomes from neuroimaging features [Park et al., 2019, Saha et al., 2018]. However, accurate interpretation of variable importance measures must be considered because they do not necessarily mean causal relationships between features and outcomes [Gillies et al., 2016]. Saha et al. [2018] used random forest to classify breast cancer subtypes and obtained AUC values up to 0.697 for Luminal A and 0.654 for triple-negative breast cancer. In another study, random forests were found to have decent but lower performance (AUC = 0.73) compared to XGBoost in predicting survival outcomes in BCBM [Li et al., 2023]. While the random forest was effective, XGBoost outperformed it, suggesting that it may be less powerful in complex survival prediction tasks. Nonetheless, random forest are still highly preferred for its stability and its ability to rank feature importance. Moreover, it was shown that random forest models achieved high accuracy in differentiating benign and malignant breast lesions using multiparametric DWI [Kallah-Dagadu et al., 2025]. Park et al. [2019] used five machine learning algorithms, random forest (RF), artificial neural network (ANN), SVM, decision tree, and naïve Bayes in comparison to logistic regression to predict key breast cancer characteristics. Their study involved 723 low-dose perfusion CT scans from 241 patients with invasive breast cancer. Eighteen neuroimaging features related to tumor blood flow and enhancement were extracted and used to predict several clinical outcomes, including lymph node status, tumor grade, ER/PR expression, HER2 status, Ki67 levels, and molecular subtypes Park et al. [2023]. Among the five algorithms, the random forest model performed best. It achieved 82% accuracy (AUC = 0.88) for ER status, 83% (AUC = 0.88) for HER2 status, and 66% (AUC = 0.82) for identifying molecular subtypes. On average, the random forest model improved classification accuracy by 13% and AUC by 0.17. Similar to SVM, RF applications in BCBM have almost exclusively focused on imaging features, limiting biological knowledge into tumor progression Park et al. [2019].

2.5.4 Extreme Gradient Boosting

Extreme Gradient Boosting (XGBoost) is a boosting-based ensemble learning method that has gained popularity in many machine learning applications [Angelone et al., 2024, Carvalho et al., 2019, Chen and Guestrin, 2016]. Unlike random forest, which builds decision trees independently, boosting algorithms construct trees sequentially so that each new tree focuses on correcting the errors of the previous ensemble Chen and Guestrin [2016]. XGBoost incorporates several improvements over traditional boosting algorithms, including regularisation, efficient tree construction, and parallel computation. These properties allow XGBoost to achieve strong predictive performance while controlling overfitting [Chen and Guestrin, 2016]. In biomedical research, XGBoost has been applied to various classification problems involving high-dimensional datasets, including cancer prognosis and imaging analysis. Furthermore, XGBoost have received increasing attention because of their high accuracy and ability to handle missing data. For instance, Erickson et al. [2017] used XGBoost to predict survival in breast cancer brain metastasis patients and achieved AUC values above 0.80 across multiple time points. Recent studies have looked into the use of machine learning to improve survival prediction in patients with BCBMAltmann et al. [2010], Park et al. [2023]. Several machine learning models were developed using data from the Surveillance, Epidemiology, and End Results (SEER) a database which provides information on cancer statistics in the United states of America (USA) described in Hayat [2007] to predict patient survival at different time points (6 months, 1 year, 2 years, and 3 years). In a study by [Li et al., 2023], six predictive models were developed namely XGBoost, logistic regression, random forest, support vector machine (SVM), k-nearest neighbor (KNN), and ID3 decision tree. Out of the six, XGBoost had the highest model perfomance, with area under the curve (AUC) values from 0.800 to 0.824 and up to 0.936 for 3-year survival [Li et al., 2023]. An example of unsupervised machine learning includes deep learning techniques such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and denoising autoencoders, which have also been applied in radiogenomic studies of breast cancer for feature extraction and classification [Goodfellow et al., 2016, Singh et al., 2020, Zhu et al., 2019]. These models have been used for prediction tasks such as multiclass classification of breast cancer subtypes. Even though, CNNs and RNNs have a strong performance, they are normally difficult to interpret and and also require large, high-dimensional datasets. For instance, Goodfellow et al. [2016] used a denoising autoencoder to extract deep neuroimaging features from breast MRI data to predict tumor features. Apart from their high predictive power, deep learning models often act as "black boxes" with limited interpretability, which may raise concerns about their clinical application [Angelone et al., 2024].

These methods each address a different statistical challenge associated with radiogenomic datasets while complementing one another. LASSO logistic regression addresses high dimensionality and multicollinearity through ℓ_1-penalised variable selection, which simultaneously shrinks irrelevant coefficients to zero and retains only the most informative predictors [Tibshirani, 1996, ?]. Support vector machines handle non-linear decision boundaries in small, high-dimensional samples through kernel-based feature mapping [Cortes and Vapnik, 1995, Schölkopf and Smola, 2002]. Ensemble methods namely random forest and XGBoost, reduce variance and capture complex feature interactions through bootstrap aggregation and sequential boosting, respectively [Breiman, 2001, Chen and Guestrin, 2016]. Together, these methods form a statistically coherent modelling framework that directly addresses the key challenges of radiogenomic analysis: high dimensionality, multicollinearity among correlated imaging predictors, and the elevated risk of overfitting in datasets where the number of predictors exceeds the number of observations [Babyak, 2004, Harrell et al., 2001].

Together, these approaches formed a coherent modelling framework. This framework addressed key challenges in radiogenomic analysis, including high dimensionality, multicollinearity among imaging predictors, and the risk of overfitting when the number of predictors exceeded the number of observations [Babyak, 2004, Harrell et al., 2001]. However, existing BCBM classification studies have primarily applied these methods to neuroimaging features alone, without evaluating whether dimensionality reduction yields features that meaningfully reflect underlying biological or genetic processes. [Fan et al., 2020, Li et al., 2023, Saha et al., 2018]. This produced models that achieved acceptable discriminative performance but offered limited biological interpretability. To address this, this study applied LASSO logistic regression to all neuroimaging features enabling simultaneous dimension reduction and variable selection [Friedman et al., 2010, Tibshirani, 1996]. Building on this, differentially expressed genes (DEGs) associated with BCBM were integrated with neuroimaging-derived physical characteristics (NIDPs). This integration restricted the neuroimaging feature space to features potentially associated with genomically influenced biological processes in metastatic brain regions [Bledsoe and Gamazon, 2024, Taha et al., 2025]. In doing so, this study aimed not only to improve classification performance but also to develop models whose selected features were biologically grounded, clinically interpretable, and reflective of the mechanisms driving BCBM.

3. Methodology

This chapter describes the methodology used for the objectives mentioned in chapter 1. The steps include data preprocessing and class balancing, LASSO-based feature selection and dimensionality reduction, genomically informed (DEG-based) feature selection and model development using multiple machine learning classifiers

3.1 Data Sources

This study used two publicly available datasets: (1) a neuroimaging dataset consisting of neuroimaging features derived from T2-weighted MRI by Taha et al. [2025], and (2) RNA sequencing gene expression dataset containing genomic profiles relevant to primary breast cancer and BCBM [Edgar et al., 2002b]. The neuroimaging dataset includes neuroimaging features extracted from pre-processed MRI scans, while the gene expression data consists of gene expression measurements obtained from RNA sequencing (RNA-seq) or microarray platforms using paired tumor samples from primary breast cancer and corresponding BCBM samples.

3.1.1 Gene Expression Data

Gene expression data was obtained from the Gene Expression Omnibus (GEO) database, which is a public resource for gene expression data. The data was processed and collected as explained by [Bos et al., 2009, Zwanenburg et al., 2020b]. The work utilised the GSE125989 dataset generated using the Affymetrix Human Genome U133A 2.0 Array platform (GPL571) [Edgar et al., 2002b]. It contains 16 matched pairs of primary breast cancer tumors and their corresponding brain metastases. Expression was measured across 22277 probe sets. Thus we have the number of variables $p \approx 22000$ being more than sample size ($n = 32$). The dataset can be accessed at `https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE125989`

3.1.2 Neuroimaging Data

The neuroimaging data was obtained from The Cancer Imaging Archive (TCIA), which can be accessed at: `https://www.cancerimagingarchive.net`. It consisted of contrast-enhanced T1-weighted (T1CE) images [Taha et al., 2025] with a sample size of ($n = 165$) and 107 neuroimaging features for each segmented region of interest (ROI), following the Image Biomarker Standardization Initiative (IBSI) guidelines [Gillies et al., 2016]. While 107 radiomic features seems a bit smaller than the sample size, the ratio p/n remains statistically insufficient for unregularized high-dimensional learning.

3.2 Image Data processing and Work Flow

Neuroimaging features extracted from MRI images form a high-dimensional feature space that can be categorised into first order statistical features, geometric, and textural properties of tumor features.

3.2.1 First-Order Statistical Features

First-order features describe the distribution of voxel intensities within a ROI, without considering spatial relationships. A voxel (volumetric pixel) is the three-dimensional equivalent of a pixel, representing a single unit of volume in a medical image [Haralick et al., 2007]. Each voxel carries an intensity value that reflects the tissue property being measured in T1-weighted MRI [Zwanenburg et al., 2020b].

Let $X = \{x_1, x_2, \ldots, x_n\}$ denote the set of voxel intensities. Common first-order statistics include:

- **Mean:** $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$
- **Variance:** $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$
- **Skewness:** measures asymmetry of the distribution
- **Kurtosis:** measures tail heaviness and peakedness

These features summarise the marginal distribution of intensities and provide information about tumor heterogeneity and tissue composition.

3.2.2 Shape-Based Features

Shape features quantify the geometric properties of tumor regions. These are independent of intensity values and depend solely on the spatial configuration of the segmented region. The following are examples:

- **Volume:** total number of voxels within the ROI
- **Surface area:** boundary extent of the tumor
- **Sphericity:** a measure of how closely the shape resembles a sphere

These features capture tumor morphology and irregularity, which may enhance class separability.

3.2.3 Texture Features

These are derived from matrices such as the Gray-Level Co-occurrence Matrix (GLCM). For example, the GLCM $P(i, j)$ represents the probability of observing intensity levels i and j at a given spatial offset. Common texture measures are contrast, correlation and homogeneity. These features capture second-order statistical dependencies and are particularly useful for characterising tumor heterogeneity.

3.2.4 Image Classification Work Flow

The overall neuroimaging analysis workflow is summarised in figure 3.1. This includes image acquisition, tumor segmentation, feature extraction, and machine learning modelling.

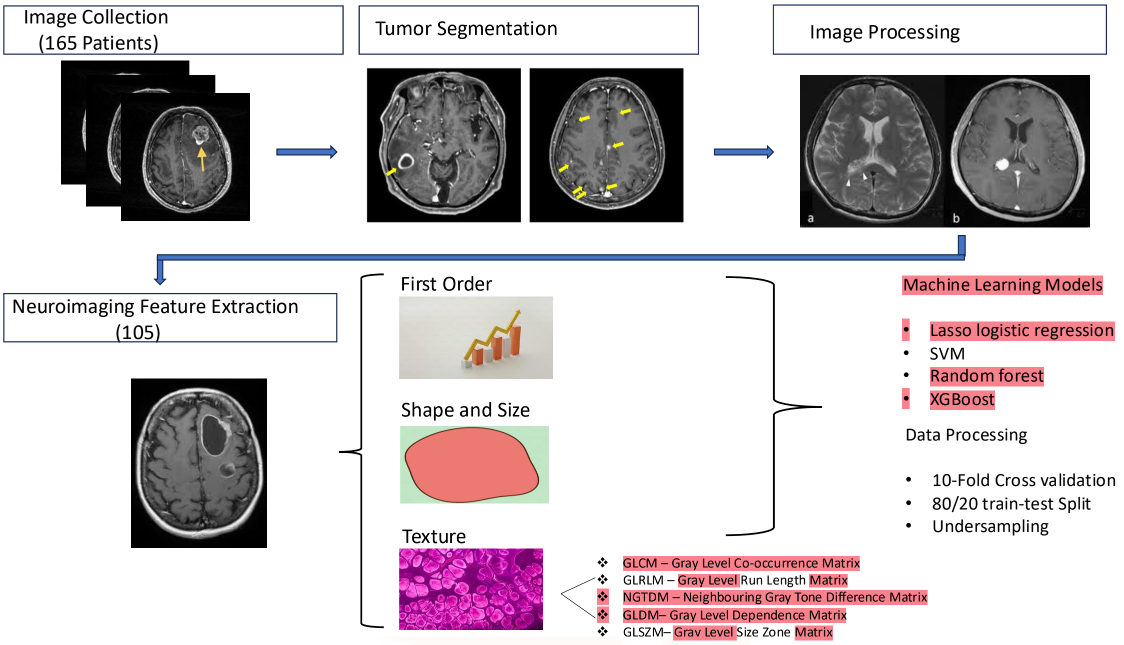

Figure 3.1: Neuroimaging Features Extraction

3.2.5 Ethical Considerations

All datasets used in this study are publicly available and contain fully anonymised patient data. The TCIA imaging datasets and GEO gene expression datasets are distributed for research purposes and comply with ethical guidelines for data sharing and patient privacy [Edgar et al., 2002a]. No identifiable patient information was accessed and no new patient data were collected; therefore, additional institutional ethical approval was not required for this study.

3.2.6 Sampling Strategy and Data Partitioning

The full neuroimaging dataset contained 530 BCBM cases and 2,295 primary breast cancer cases. Let n denote the total number of observations. To address class imbalance, random undersampling of the majority class was performed [He et al., 2024]. Let n_1 and n_0 denote the number of observations in the minority (BCBM) and majority (primary) classes, respectively, with $n_0 > n_1$. A subset of size n_1 was randomly sampled from the majority class to build a balanced dataset of size $2n_1 = 1,060$ observations (530 BCBM and 530 primary cases). This balanced dataset ($n = 1,060$) was then divided into training and testing subsets using an 80:20 split. This was such that $0.8n = 848$ observations were used for model training and $0.2n = 212$ observations were for independent evaluation [Ruppert, 2004]. Similarly, the genom-identified dataset, restricted to DEG-associated brain regions, contained 304 BCBM cases and 678 primary breast cancer cases. Applying the same undersampling procedure resulted in a balanced dataset of size $2n_1 = 608$ (304 BCBM and 304 primary cases). This dataset was split into $0.8n = 486$ training and $0.2n = 122$ testing. To further improve the robustness of the models, repeated k-fold cross-validation was used. Specifically, the training data were partitioned into $k = 10$ folds, and the cross-validation procedure was repeated multiple times. This approach provided a more stable estimate of model performance [Park et al., 2019].

3.3 Identification of Differentially Expressed Genes

3.3.1 Linear Models for Microarray Analysis (LIMMA)

To identify genes that are significantly associated with BCBM, differential expression analysis was performed on the gene expression dataset (GSE125989) using Linear Models for Microarray Analysis (LIMMA) as described by [Smyth et al., 2003]. LIMMA was chosen due to its empirical Bayes approach, which improves variance estimation and statistical power, particularly in studies with small sample sizes [Taha et al., 2025]. Using this approach, genes that are significantly up- or down-regulated in BCBM relative to primary breast cancer were identified as DEG's. LIMMA fitted a separate linear model to each gene across all samples simultaneously, analysing the entire experiment as an integrated whole rather than making pairwise comparisons between groups [Ritchie et al., 2015]. For gene g, the model is:

$$E(\mathbf{y}_g) = \mathbf{X}\boldsymbol{\alpha}_g, \tag{3.1}$$

where:

- $\mathbf{y}_g$ is the vector of $\log_2$-transformed expression values for gene g across all n samples;
- $\mathbf{X}$ is the design matrix encoding the experimental groups, specifically primary breast tumor and BCBM;
- $\boldsymbol{\alpha}_g$ is the vector of gene-specific coefficients estimated through ordinary least squares [Harrell et al., 2001, Smyth et al., 2003] such that:

$$\hat{\boldsymbol{\alpha}}_g = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}_g. \tag{3.2}$$

- The gene-wise residual variance is estimated as:

$$s_g^2 = \frac{(\mathbf{y}_g - \mathbf{X}\hat{\boldsymbol{\alpha}}_g)^T(\mathbf{y}_g - \mathbf{X}\hat{\boldsymbol{\alpha}}_g)}{d_g}, \tag{3.3}$$

where:

- $d_g = n - \text{rank}(\mathbf{X})$ is the residual degrees of freedom, which in this study equals $d_g = 30$, since $n = 32$ samples and $\text{rank}(\mathbf{X}) = 2$ corresponding to the two groups (primary breast cancer and BCBM)
- the log-fold-change (logFC) for each gene, comparing BCBM to primary, is obtained directly from the estimated contrast of interest in $\hat{\alpha}_g$, where a positive logFC indicates higher expression in BCBM (upregulation) and a negative logFC indicates lower expression in BCBM (downregulation).

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3.3.2 Empirical Bayes Variance Moderation

With only 32 samples, gene-wise variance estimates s_g^2 are unreliable because they are based on few degrees of freedom. LIMMA addresses this through empirical Bayes shrinkage, which moderates each gene-wise variance estimate toward a global prior variance estimated from the full ensemble of all genes on the array [Smyth et al., 2003]. The prior distribution on the gene-wise variances is:

$$\frac{1}{\sigma_g^2} \sim \frac{1}{d_0 s_0^2}\chi^2_{d_0}, \tag{3.4}$$

where:

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- s_0^2 is the prior variance estimated from the data across all genes;

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- d_0 is the prior degrees of freedom, also estimated from the data across all genes.

The moderated posterior variance estimate $\tilde{s}_g^2$ is then obtained by combining the gene-wise estimate s_g^2 with this prior:

$$\tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g}, \tag{3.5}$$

where:

- $d_0 + d_g$ are the posterior degrees of freedom, which are greater than d_g alone, reflecting the additional information borrowed from the ensemble of all genes [Smyth et al., 2003]
- this shrinkage pulls unreliable gene-specific variance estimates toward the global average [Ritchie et al., 2015].

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The moderated t-statistic for testing differential expression of gene g is:

$$\tilde{t}_g = \frac{\hat{\alpha}_g}{\tilde{s}_g \sqrt{v_g}}, \tag{3.6}$$

where:

- $\hat{\alpha}_g$ is the estimated log-fold-change for the contrast of interest, obtained from the least squares solution in Equation (3.2).
- v_g is the variance factor from the linear model associated with the contrast of interest.
- $\tilde{s}_g = \sqrt{\tilde{s}_g^2}$ is the moderated posterior standard deviation from Equation (3.5).

Under the null hypothesis of no differential expression, $\tilde{t}_g$ follows a t-distribution with $d_0 + d_g$ posterior degrees of freedom. This is greater degrees of freedom than the ordinary gene-wise t-statistic would have, which directly improves statistical power to detect differentially expressed genes in small samples [Huang et al., 2009, Ritchie et al., 2015].

3.4 Dimension Reduction Techniques

This section describes the two methods to select variables in order to reduce the dimension of the image data for classification. The study contrast LASSO, which enforces sparsity through coefficient shrinkage, with a radiogenomic approach, which leverages in silico infusions between transcriptomic data to restrict only neurological features that are biologically relevant from the neuro-anatomical regions.

3.4.1 LASSO Logistic Regression for Variable Selection

The first method applied for feature selection from the 165 observations, each described by 107 neuroimaging features. LASSO logistic regression was applied to the reduced feature set $\mathcal{F}^*$ to perform simultaneous parameter estimation and variable selection [Tibshirani, 1996]. LASSO introduces an ℓ_1 penalty to the log-likelihood function, which shrinks the coefficients of less relevant features toward zero, effectively removing them from the model and retaining only the most informative predictors [Friedman et al., 2010]. The logistic regression model is given by:

$$P(Y = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-\mathbf{x}_i^T \boldsymbol{\beta})}, \tag{3.7}$$

where:

- $Y \in \{0, 1\}$ is the binary outcome variable, where $Y = 1$ indicates the presence of a BCBM tumor and $Y = 0$ indicates a primary breast tumor;
- $\mathbf{X} = (X_1, X_2, \ldots, X_p)^T$ is the p-dimensional vector of predictor variables;

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- $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T$ is the observed feature vector for patient i;

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- $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^T$ is the vector of unknown model coefficients, where β_0 is the intercept;
- $\mathbf{x}_i^T \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ is the linear predictor.

280 The logistic (sigmoid) function ensures that predicted values lie between 0 and 1, allowing them to be interpreted as probabilities [Hosmer Jr et al., 2013].

105 For interpretability, the model can be expressed in terms of the log-odds (logit):

$$\log \left(\frac{P(Y = 1 \mid \mathbf{x}_i)}{1 - P(Y = 1 \mid \mathbf{x}_i)} \right) = \mathbf{x}_i^T \boldsymbol{\beta}. \tag{3.8}$$

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9 This formulation shows that while the probability of BCBM is a non-linear function of the predictors, the log-odds are linearly related to the feature vector. Each coefficient β_j represents the change in the log-odds of BCBM associated with a one-unit increase in predictor X_j, holding all other variables constant. This modelling framework is consistent with standard logistic regression approaches [Hosmer Jr et al., 2013, Ruppert, 2004].

Regularisation with LASSO

43 Logistic regression with LASSO (L1) is a widely used method in statistics and machine learning for estimating relationships between variables and classifications. 45 [Loh, 2011, Tibshirani, 1996]. L1 is the sum of the absolute values of the coefficients [Chicco and Jurman, 2021]. The main aim is to balance model simplicity with accuracy. This method works well for high-dimensional data, like radiogenomic features, where there are more predictors than observations. LASSO imposed an L1 190 regularization penalty to perform feature selection by shrinking less relevant coefficients to zero, ultimately identifying the most discriminative features associated with BCBM and the likelihood of developing the condition [Tibshirani, 1996].

The LASSO regression model was fitted to the binary classification task of predicting the primary breast cancer or BCBM. The objective function optimized during the LASSO regression was:

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$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{N} \sum_{i=1}^{N} [-y_i \log(p_i) - (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \tag{3.9}$$

where:

- $p_i = \frac{1}{1+\exp(-\mathbf{x}_i^T \boldsymbol{\beta})}$ is the logistic function.
- $y_i \in \{0, 1\}$ is the label that indicates BCBM.

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- $\mathbf{x}_i$ is the vector for patient i.

- N is the total number of patients.
- p is the number of neuroimaging features.
- λ is the regularization hyperparameter controlling sparsity.
- β_j is the regression coefficient associated with the j-th predictor (feature)

This method was implemented using the `glmnet` package in R Friedman et al. [2010], with 10-fold cross-validation to select the optimal value of λ.

3.4.2 Genomic Guided Dimensionality Reduction

The second approach used in this study involves a genomic guided approach where genes identified in section 3.3 are used to identify phenotypic features and hence inform variable selection.

In Silico Radiogenomics Intergration

Although it is possible to link the gen data (section 3.1.1) and image data (section 3.1.2), one major statistical challenge in this study is the lack of a single prospective cohort containing both RNA sequencing and neuroimaging data for the same patients. Thus, the work adopted an in-silico radiogenomic fusion where the GEO dataset (GSE125989) serves as the major discovery cohort that identifies statistically significant genomic signatures (genes) of metastasis via differential expression analysis. These significant genes are then used to inform feature selection in the TCIA data set which can be needed fro BCBM tumor detection through a classification model. Thus we are able to evaluate on whether the biological drivers like brain functionality, structure, neurological effect , etc identified by the transcriptomic space can effectively guide selection of MRI features in a high dimensional space through linking these datasets clinical phenotype.

Correlation Analysis: DEGs and Neuroimaging-Derived Phenotypes

To explore the association between gene expression and neuroimaging features, the study made use of the `NeuroimaGene` R package [Taha et al., 2025]. This package identified any correlations between genetically regulated gene expression (GRex) and neuroimaging-derived phenotypes (NIDPs) with stronger correlations suggesting that specifc imaging features may show activity of the genes associated with metastasis [Bledsoe and Gamazon, 2024]. The study used GReX and NIDPs from the UK Biobank; a reference for the mapping of brain structure and function effects [Bycroft et al., 2018]. Primarily, the package's function was to identify neuroimaging-derived brain features that are affected by genetically regulated expression of user-provided genes or gene sets. Additionally, it identified brain regions where the genetically regulated expression of a given gene affected is relevant neurologically [Bledsoe and Gamazon, 2024]. A set of DEG's that are associated with neuroimaging features were produced by this process which were used to deepen our understanding of BCBM and to support further discriminative modelling [Bledsoe and Gamazon, 2024].

Let g_k denote the genetically regulated expression level of gene k, and let z_m denote a neuroimaging-derived phenotype associated with a particular brain region. The association between gene expression and the neuroimaging phenotype is modelled as a simple linear regression [Harrell et al., 2001]:

$$z_m = \alpha + \gamma g_k + \varepsilon, \tag{3.10}$$

where:

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- α is the intercept, representing the expected value of the imaging phenotype when genetically regulated expression is zero.

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- γ is the regression coefficient measuring the strength and direction of the association between genetically regulated gene expression and the neuroimaging phenotype, with a statistically significant γ indicating that variation in g_k explains variation in z_m at a population level.

113 221
- $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is the random error term, assumed to be independently and identically distributed with mean zero and constant variance σ^2.

Statistical significance of each gene-phenotype association was assessed after applying a Bonferroni correction for multiple comparisons across all genes and NIDPs tested [Bledsoe and Gamazon, 2024]. Significant associations identify brain regions whose structural or functional properties are influenced by genes associated with BCBM. These regions were used to guide the construction of the DEG-associated neuroimaging feature set for classification modelling, as described in Section 3.5. 92

The full set of 107 neuroimaging features was reduced to a genomically informed subset by retaining only those features derived from brain regions significantly associated with BCBM-related differentially expressed genes, as identified through the GReX-NIDP correlation analysis described in Section 3.3 [Bledsoe and Gamazon, 2024, Mazurowski, 2015]. This approach addresses high dimensionality from a biological perspective. Specifically, features are selected not only based on their association with the outcome but also on their biological relevance, as they originate from brain regions whose structural and functional properties are genomically linked to BCBM [Fan et al., 2020].

This results in a reduced feature set $\mathcal{F}^* \subseteq \mathcal{F}$, where $\mathcal{F} = \{f_1, f_2, \ldots, f_{107}\}$ represents the full set of neuroimaging features extracted from MRI images across all 165 patients. The subset $\mathcal{F}^*$ contains only those features whose associated brain regions show significant relationships with at least one differentially expressed gene.

3.5 Machine Learning Classification Models

To address Objectives 1, 2, and 5, four classification models were developed namely LASSO logistic regression, SVM, random Forest, and XGBoost. These models were applied to three types of data, the neuroimaging features alone comprising 107 neuroimaging features extracted from 165 patients, the 29 LASSO selected features and a combined dataset integrating neuroimaging and gene expression data. The choice of models was guided by the high-dimensional nature of the data ($p > n$), which is common in radiogenomics studies [Aerts et al., 2014, Gillies et al., 2016].

Classification Models

In the final stage, SVM, random forest, and XGBoost models were trained using the selected feature set $\mathcal{F}^{**}$ to evaluate the robustness and venerability of the features. This approach ensures that all models are assessed on the same biologically and statistically refined feature space, allowing for a comparison of classification performance. SVM was selected for its ability to model complex, non-linear relationships in high-dimensional data through kernel-based transformations [Cortes and Vapnik, 1995, Schölkopf and Smola, 2002]. Random forest was included due to its ensemble learning framework, which reduces variance by aggregating multiple decision trees and provides measures of feature importance [Breiman, 2001]. XGBoost was chosen for its boosting-based approach, which iteratively minimises a regularised loss function and improves predictive performance by correcting residual errors at each step [Chen and Guestrin, 2016]. This framework addresses key challenges in the data by reducing high dimensionality, enhancing biological interpretability, and improving model stability through the use of regularisation and ensemble learning methods [Breiman, 2001, Ruppert, 2004].

3.5.1 Support Vector Machines

SVM is widely used in machine learning as it can handle both linear and non-linear classification and prediction tasks [Noble et al., 2004]. However, when the data are not linearly separable, kernel functions are used to transform the higher-dimensional space of the data to enable linear separation Carvalho et al. [2019]. They distinguish between two classes by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes [Teja et al., 2024]. Given the high dimensionality and complex non-linear relationships between imaging features and genomic markers, kernel-based learning methods become highly relevant. Significant advancements were made including the development of the soft margin SVM (non-linear) described in Altmann et al. [2010], which allows for better handling of nonseparable data. For this study, a nonlinear

SVM with a Radial Basis Function (RBF) kernel was employed, defined as

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$

where γ controls the influence of individual training samples. The RBF kernel was selected because neuroimaging and genomic features exhibit complex, non-linear interaction patterns that are not captured by a linear kernel, and the RBF kernel is well-suited to high-dimensional spaces where the relationship between classes cannot be approximated by polynomial boundaries of fixed degree [Noble et al., 2004]. It was used for both prediction and explanation of classification attributes.

The decision function is defined as:

Given a training dataset (x_i, y_i) where $y_i \in \{-1, 1\}$, the optimisation problem is

$$\min_{w,b} \frac{1}{2}||w||^2 + C \sum_{i=1}^{n} \xi_i$$

subject to

$$y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i$$

where

- w is the normal vector to the hyperplane,
- b is the intercept,
- C controls the trade-off between margin size and classification error,
- ξ_i are slack variables,
- $\phi(x)$ is a feature mapping function.

The SVM classified tumor regions based on neuroimaging and genomic features, also helped to identify the features that contributed the most to the separation margin, and potentially linked these features to specific regions of the brain. SVMs are implemented using the `e1071` package in R (version 4.0.2) Meyer et al. [2019].

3.5.2 Random forest

Random forest is a commonly used machine learning algorithm that combines the output of multiple decision trees to reach a single result [Breiman, 2001]. Although decision trees are common supervised learning algorithms, they can be prone to problems such as bias and overfitting. However, when multiple decision trees form an ensemble in the random forest algorithm, they predict more accurate results, particularly when the individual trees are not correlated with each other [Breiman, 2001]. Random forests incorporate a bootstrapping strategy along with additional layers of randomness and tuning mechanisms Orrantia-Borunda et al. [2022].

Tree Construction and Bootstrapping

Let the training dataset be $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, where $x_i \in \mathbb{R}^p$ denotes the feature vector for observation i (comprising neuroimaging and genomic features) and $y_i \in \{0, 1\}$ denotes the class label. For each tree $b = 1, \ldots, B$, a bootstrap sample $\mathcal{D}^{*b}$ of size n is drawn with replacement from $\mathcal{D}$. Each tree T_b is then grown on $\mathcal{D}^{*b}$ by recursively partitioning the feature space. At each internal node, a random subset of $\texttt{mtry} \leq p$ features is selected, and the optimal binary split is chosen by maximising the reduction in node impurity. For classification tasks, the Gini impurity is used [Lundberg et al., 2018]:

$$G = \sum_{k=1}^{K} \hat{p}_k (1 - \hat{p}_k)$$

where $\hat{p}_k$ is the proportion of observations in the node belonging to class k, and K is the number of classes. A split is chosen at threshold s on feature j such that

$$\Delta G = G(\text{parent}) - \frac{n_L}{n} G(\text{left}) - \frac{n_R}{n} G(\text{right})$$

is maximised, where n_L and n_R are the number of observations in the left and right child nodes respectively. Each tree is grown until a stopping criterion is met, typically when a node contains fewer than `nodesize` observations, at which point it becomes a terminal (leaf) node assigned the majority class of its members [Lundberg et al., 2018].

Ensemble Aggregation

The ensemble prediction aggregates across all B trees [Breiman, 2001]. For a new observation x, each tree $T_b(x)$ returns a class prediction. The random forest classifier is defined as the majority vote:

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^{B} T_b(x)$$

where B is the total number of trees in the forest and $T_b(x) \in \{0, 1\}$ is the class predicted by the b-th tree for input x. The averaging over B trees reduces the variance of the individual tree predictions without increasing bias, exploiting the bias-variance decomposition [Demetriou et al., 2024]:

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Since each tree is trained on a different bootstrap sample and uses a random feature subset at each split, the trees are decorrelated. This ensures that the variance of

254 the ensemble decreases approximately as σ^2/B as $B \to \infty$, where σ^2 is the variance of a single tree Lundberg et al. [2018].

Tuning Parameters

A random forest model can be controlled through several key parameters as follows [Breiman, 2001]:

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- `ntree` (B): The number of trees in the forest. A larger B reduces variance but increases computational cost. The model error typically stabilises beyond a sufficient number of trees, after which additional trees yield diminishing returns.

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- `sampsize`: The number of observations sampled to train each individual tree. In practice, this is often set equal to n, particularly when sampling with replacement (bootstrapping).

13
- `mtry` (m): The number of randomly selected features considered for splitting at each internal node, where $m \leq p$. The default for classification is $m = \lfloor\sqrt{p}\rfloor$. A larger `mtry` is preferred for sparse signals; a smaller `mtry` is preferred when features are highly correlated, as it encourages greater tree diversity.

207 144
- `nodesize`: The minimum number of observations required in a terminal node before splitting is halted. Smaller values allow deeper trees and lower bias but higher variance; larger values act as a regularisation mechanism, analogous to the smoothing parameter k in k-nearest neighbours [Breiman, 2001].

Random Forest as a Kernel Estimator

Similar to a tree model, random forest can be interpreted as a kernel estimator [Schölkopf and Smola, 2002]. The proximity between two observations x and x' is defined as the proportion of trees in which both observations fall into the same leaf node:

220 63 63

$$K_{\mathbf{RF}}(x, x') = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\left[\ell_b(x) = \ell_b(x')\right]$$

where $\ell_b(x)$ denotes the leaf node assigned to x by tree b, and $\mathbf{1}[\cdot]$ is the indicator function [Lundberg et al., 2018]. This implicitly defines a proximity-based similarity metric that accounts for interactions between neuroimaging texture features and genomic expression patterns. The random forest model provided robust classification of primary breast cancer and BCBM. Feature importance scores were computed using the mean decrease in Gini impurity across all trees, indicating which neuroimaging and genomic attributes were most discriminative. This strengthened the identification of important features and brain regions associated with tumor characteristics [Erickson et al., 2017]. Random forests were implemented using the `randomForest` package in R [Breiman, 2001].

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3.5.3 Extreme Gradient Boosting

Boosting is another ensemble model. A more general and statistically motivated approach called Gradient Boosting Machines(GBM), introduced by Arefan [2020], extends boosting to work with any differentiable loss function. XGBoost builds trees sequentially, minimizing the regularized loss function. A speed and performance-optimized boosting algorithm Chen and Guestrin [2016].

The objective function is defined as:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where

- $l(y_i, \hat{y}_i)$ is the loss function,
- f_k represents the k-th decision tree,
- $\Omega(f_k)$ is a regularisation term controlling model complexity.

This XGboost offered interpretable neuroimaging feature importance through built-in gain and SHAP (SHapley Additive exPlanations) values Rozemberczki et al. [2022], helping in the identification of the most influential features.

3.6 Data Preprocessing and Cross-validation

3.6.1 Neuroimaging Data Cleaning and Transformation

The Classification analysis in this section utilises three data sets to compare the effectiveness of the two variable selection methods. The three data sets used are described as follows:

1. Original data set neuroimaging dataset in section (3.1.2) $(n = 165,\ p = 107)$
2. LASSO reduced data set: reduced p due to regularisation as described in section 3.4.1 $(n = 165,\ p < 107)$
3. Genomic reduced data set: reduced p due to biological importance as described in section 3.4.2 $(n = 165,\ p < 107)$

3.6.2 Parameter Tuning and Classification Model Optimisation

LASSO Logistic regression: The LASSO model used an L1 penalty with the regularisation parameter λ selected through 10-fold cross-validation. The minimum value of λ was considered as the final classification model.

Support Vector Machine: The SVM model with radial basis function kernel was tuned using a grid search over the cost parameter C and kernel width parameter σ. The candidate grid included:

- $C \in \{0.1, 1, 10\}$
- $\sigma \in \{0.01, 0.05, 0.1\}$

The optimal hyperparameter combination was selected based on maximisation of cross-validated AUC. The final SVM model achieved its best performance using the selected (C, σ) pair identified during repeated 10-fold cross-validation.
Random forest The random forest model was trained using an ensemble of 500 trees ($ntree = 500$). The number of variables randomly selected at each split ($mtry$) was tuned using cross-validation over the following candidate values:

- $mtry \in \{2, 4, 6, 8\}$

ptimal $mtry$ value was selected based on the highest cross-validated AUC, ensuring optimal balance between bias and variance. XGBoost The XGBoost model was trained using a gradient boosting framework with logistic loss. The following hyperparameter were specified:

- Maximum tree depth: $max\_depth = 3$
- Learning rate: $\eta = 0.1$
- Subsampling rate: 0.8
- Column subsampling rate: 0.8

The optimal number of boosting rounds was determined using 10-fold cross-validation with early stopping, resulting in a data-driven selection of $nrounds$ that maximised AUC while preventing overfitting.

3.6.3 Addressing Class Imbalances

The neuroimaging dataset in section(3.1.2) had class imbalance between patients with primary breast cancer and those with BCBM. The full neuroimaging dataset contained 530 BCBM cases and 2,295 primary breast cancer cases. The genom-identified dataset, restricted to DEG-associated brain regions, contained 304 BCBM cases and 678 primary breast cancer cases. To address this imbalance, several undersampling strategies were explored using the `smotefamily` package in R version 4.5.0 [de Micheaux et al., 2014]. These included random undersampling at different ratios (1:1 and 1:2), systematic undersampling, and cluster-based undersampling of the majority class. Random undersampling with a 1:1 ratio (equal numbers of BCBM and primary breast cancer cases) provided the most stable and balanced performance across both the full neuroimaging dataset and the genom-identified dataset. This method was therefore selected for the final analysis.

3.7 Model Evaluation

Model performance was evaluated using classification metrics derived from the confusion matrix, a fundamental tool for both performance assessment and model interpretability [Sathyanarayanan and Tantri, 2024]. The confusion matrix provides a comparison between predicted and actual class labels, enabling information into how a model makes correct and incorrect classes [Sathyanarayanan and Tantri, 2024]. Additionaly, it provides a class-wise breakdown of errors, making it particularly valuable for interpreting model behaviour in clinical applications [Carvalho et al., 2019]. These metrics quantify the discriminative ability of the model and are widely used in radiogenomics studies, where both predictive accuracy and clinical reliability are essential [Chicco and Jurman, 2021, Severn et al., 2022].

3.7.1 Performance Metrics

The analysis considered a binary outcome variable $Y \in \{0, 1\}$, where $Y = 1$ indicates the presence of BCBM and $Y = 0$ indicates absence. For each observation with neuroimaging features $X = x$, the model produces a predicted label $\hat{Y} \in \{0, 1\}$. The confusion matrix is composed of four fundamental components: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). These provide a complete representation of classification outcomes and form the basis for several evaluation metrics [Fan et al., 2020].

The performance metrics are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{3.11}$$

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN}, \tag{3.12}$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{3.13}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{3.14}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \tag{3.15}$$

Each metric shows a different aspect of classification performance. Sensitivity quantifies the ability to correctly classify BCBM, while specificity reflects the ability to correctly identify non-tumor cases [Sathyanarayanan and Tantri, 2024]. Precision measures the reliability of positive classes, and the F1-score provides a balance between precision and sensitivity [Agarwal et al., 2021, Naidu et al., 2023]. Owings to the fact that accuracy may be misleading in imbalanced datasets, which are common in medical imaging, multiple metrics are reported to provide a more comprehensive assessment [Chicco and Jurman, 2021, Saito and Rehmsmeier, 2015].

To evaluate performance across all possible thresholds, receiver operating characteristic (ROC) analysis was used. The ROC curve plots sensitivity against the false positive rate (1 − specificity). The area under the ROC curve (AUC) provides a threshold-independent measure of discrimination and is widely used in radiomics studies to assess model separability [Fawcett, 2006, Hanley and McNeil, 1982].

3.7.2 Model Interpretability

In addition to performance, model interpretability is essential for clinical applicability in radiomics. Neuroimaging models are usually based on high-dimensional features extracted from medical images, which can reduce transparency and further limit climical trust in model classifications [Cama et al., 2025, Severn et al., 2022, Singh et al., 2020]. In this study, model interpretability was assessed using regression coeeficients for LASSO logistic regression model,feature importance measures and SHAP for tree based models and support vector machine.

Regression Coefficients Interpretation

In LASSO logistic regression, each coefficient β_j represents the change in the log-odds of metastasis associated with a one-unit increase in predictor X_j, holding other variables constant [Hosmer Jr et al., 2013, Tibshirani, 1996]. A positive coefficient indicates an increased likelihood of metastasis, while a negative coefficient implies a decrease in the log-odds of metastasis. The magnitude of the coefficient reflects the strength of association.

SHapley Additive exPlanations

SHAP is a machine learning tool that can explain the output of any model by computing the contribution of each feature to the final classification [Lundberg et al., 2018, Lundberg and Lee, 2017]. It is game theory based method, where each feature is treated as a 'player' and the prediction is the 'payout' to be fairly distributed among all players. The SHapley value of each feature represents its average marginal contribution across all possible subsets (coalitions) of features [Lundberg et al., 2018, Lundberg and Lee, 2017].

The SHAP explanation model decomposes a prediction into a linear combination of feature contributions as shown in the equation below;

$$g(z') = \phi_0 + \sum_{j=1}^{p} \phi_j z'_j, \tag{3.16}$$

where,

- g denotes the explanation model used to approximate the original predictive model.

180 167 148

- p represents the total number of input features.
- $z' \in \{0,1\}^p$ is the coalition vector, where the j-th element z'_j indicates whether feature j is included ($z'_j = 1$) or excluded ($z'_j = 0$).
- ϕ_0 is the base value, defined as the mean model output over the training dataset.
- ϕ_j denotes the Shapley value for feature j, quantifying its contribution to the model prediction relative to the baseline.

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3.7.3 Feature Importance

Feature importance measures the contribution of each predictor variable to a model's prediction. Two standard approaches are considered here: tree-based impurity reduction (Gini importance) and permutation-based importance [Breiman, 2001].

Tree Based Feature Importance

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Gini importance quantifies the contribution of a feature as the reduction in node impurity it produces, weighted by the proportion of samples reaching that node [Altmann et al., 2010]. For a node j, the node importance ni_j is defined as:

38

$$ni_j = w_j C_j - w_{\text{left}(j)} C_{\text{left}(j)} - w_{\text{right}(j)} C_{\text{right}(j)}, \tag{3.17}$$

where:

28

- ni_j is the importance of node j;
- w_j is the weighted number of samples reaching node j, expressed as a proportion of the total number of training samples (the node probability);
- C_j is the impurity value at node j;
- left(j) and right(j) denote the left and right child nodes of node j, respectively.

59 59

The feature importance fi_i for feature i is then computed as the sum of node importances across all nodes j that split on feature i, normalised by the sum of node importances across all nodes in the tree Altmann et al. [2010]:

235

$$fi_i = \frac{\sum_{j:\,\text{node } j \text{ splits on feature } i} ni_j}{\sum_{j:\,\text{all nodes}} ni_j}. \tag{3.18}$$

This normalisation ensures that the feature importances across all features sum to one, allowing direct comparison of relative contributions [Altmann et al., 2010].

219 For a random forest of T trees, the final importance of feature i is averaged across all trees:

$$\text{Final Importance}(i) = \frac{1}{T}\sum_{t=1}^{T} fi_i^{(t)}, \tag{3.19}$$

214 where $fi_i^{(t)}$ is the normalised importance of feature i in the t-th tree [Altmann et al., 2010].

Permutation Feature Importance

31 Permutation feature importance measures how much a model's performance drops when the values of a feature are randomly shuffled, thereby breaking its relationship with the outcome [Loh, 2011]. For feature j, this is defined as:

$$FI_j = \text{Score}_{\text{baseline}} - \text{Score}_{\text{permuted}_j}, \tag{3.20}$$

290 where $\text{Score}_{\text{baseline}}$ is the model accuracy evaluated on the original validation data, and $\text{Score}_{\text{permuted}_j}$ is the model score after the values of feature j have been per-
55 muted. A large value of FI_j indicates that the model relies heavily on feature j;
55 a value near zero suggests the feature contributes little to predictive performance [Breiman, 2001] under the SVM.

This chapter outlined the methodology used to develop classification models for BCBM using neuroimaging and genomic data. It described the data sources, preprocessing, differential gene expression analysis, and the integration of DEG-selected genomic features with neuroimaging data. The chapter also presented
303 the statistical learning methods employed, including LASSO, SVM, random forest, and XGBoost, along with model training and evaluation procedures. Overall, the methodology was designed to address high-dimensional challenges while ensuring robust and interpretable modelling.

4. Results

4.1 Overview

This chapter presents the results of the analysis described in Chapter 3. The findings are organized by study objectives. Tables, figures, and summaries show the key findings across three main sections: Classification models using neuroimaging features, gene expression analysis with neuroimaging associations, and classification models using neuroimaging features selected using gene data.

4.2 Base Model Classification Results

This section presents the results for the basiline classification models trained on the full neuroimaging feature dataset as described in section 3.1.2. The focus will be how the performance of four models namely LASSO Logistic Regression, Random Forest, Support Vector Machine and XGBoost. This section also seeks to answer the first objective as outlined in section 1.1.

4.2.1 Feature Correlation and Multicollinearity

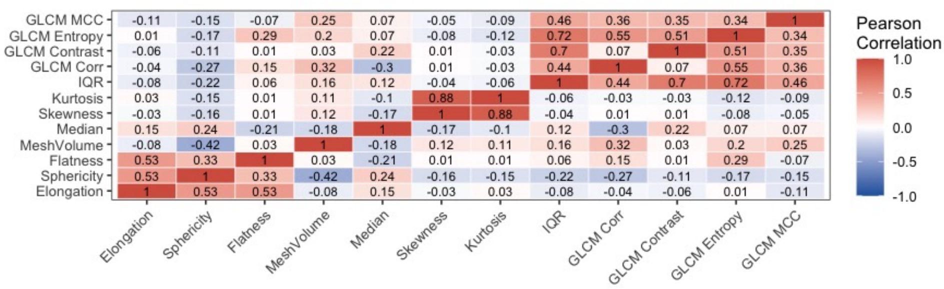

Figure 4.1: Pearson correlation heatmap of neuroimaging features

Figure 4.1 presents the Pearson correlation matrix of the extracted neuroimaging features using full data as in section 3.1.2. It show that there exist some strong positive correlations (correlations $\rho > 0.5$) specifically among shape-based features including skewness and kurtosis ($\rho = 0.88$), together with flatnenss and elongation ($\rho = 0.53$). This may indicate the shape of the brain ROI (Region of Interest) distribution becomes more positively skewed and more leptokurtic. Similarly, texture-based features derived from the Gray-Level Co-occurrence Matrix (GLCM) entropy an IQR ($\rho = 0.72$) also shows high correlations. This may indicate brain texture complexity, chaotic and random. Multicollinearity presence inflates the variance of parameter estimates and reduces model interpretability, particu-

larly in linear models such as logistic regression [Harrell et al., 2001] hence the need for penalised regression methods and ensemble-based approaches, which are more robust to correlated predictors and high-dimensional feature spaces[Chen and Guestrin, 2016, Tibshirani, 1996]. Therefore, the following four machine learning models were used in this study, as outlined in Chapter 3.

4.2.2 Baseline LASSO Logistic Regression (BLASSO)

The classification results described in this section are as a result of LASSO regression model employed on all 107 neuroimaging features from the neuroimage data in section 3.1.2.

Feature Selection:

The model used a correlation-based filtering step was applied to reduce redundancy among highly collinear features ($|r| > 0.9$). This reduced the feature space from 107 to 52 features. Additionally, LASSO logistic regression was used to further perform feature selection and regularisation using 10-fold cross-validation, the optimal regularisation parameter of λ_{min} ($= 3.99 \times 10^{-4}$) was obtained corresponding to a maximum AUC (≈ 0.735) as shown in Figure 4.2. This indicate a very week regularisation being applied. Additionally it can be seen that λ_{1SE} has a more conservative model within one standard error of the minimum. This optimisation process further reduced the feature set from 52 to 29 selected neuroimaging features. Therefore, the LASSO regularisation crtiteria managed to reduce the dimension by around 27% from the original 107 features. These 29 features will be used to analyse the effect of LASSO regularisation as explained in section 3.4.1.

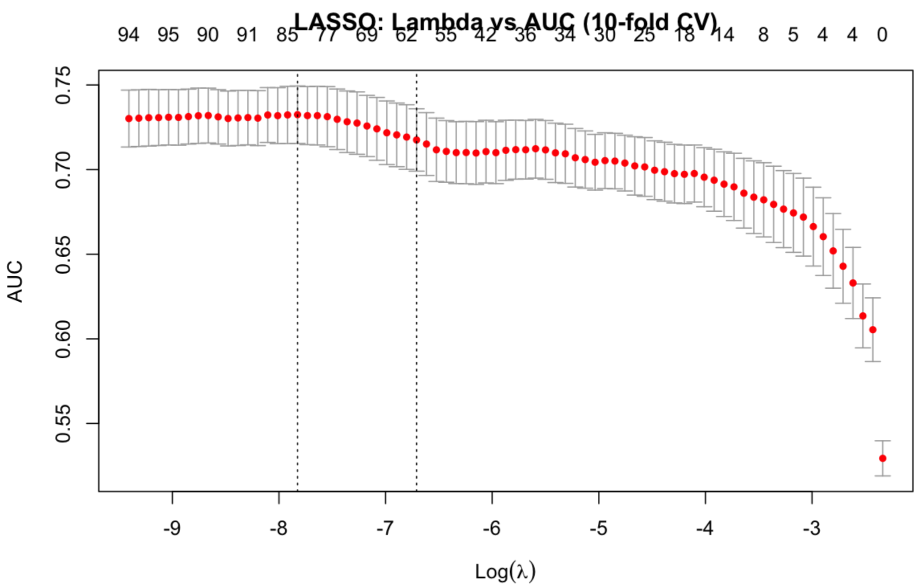

Figure 4.2: Ten-fold cross-validation curve for LASSO regularisation.

This method selected features were predominantly first-order intensity measures including: (*RootMeanSquared*, *Mean, Variance*) and texture-based features (*Gray Level Variance (GLDM)* and *Long Run Low GrayLevel Emphasis (GLRLM)* from GLRLM) as indicated by the coefficients ($\hat{\beta}$) of top five dominant features in Table 4.1. The table shows that the largest influence on BCBM tumor detection comes from *original firstorder RootMeanSquared* with a nagative coefficient of ($\hat{\beta} = -21.5$). This indicates that unit increase in this predictor will results in a decrease of log-odds of BCBM tumor by 21.5 units. However, positive coefficient values on the other hand like *original firstorder Mean* ($\hat{\beta} = 19.5$) showed a strong positive association. It means that an increase in the tumor mean intensity is associated with increased likelihood for BCBM tumor. Table 4.5 show that the LASSO Regression

Table 4.1: Top 5 features selected by LASSO Regression Model

Feature	Coefficient
original firstorder RootMeanSquared	-21.516
original firstorder Mean	19.55
original gldm GrayLevelVariance	-11.347
original firstorder Variance	10.225
original glrlm LongRunLowGrayLevelEmphasis	-8.33

model performed moderately high with AUC of 74.8%, 66.5% Accuracy and F1 score of 69.2%.

4.2.3 Basiline Random forest (BRNF)

Baseline Random forest model was applied to the full set of 107 neuroimaging features as a non-linear ensemble learning method for classification. Unlike LASSO Logistic regression model results in section 4.2.2, the major features influencing BCBM tumor detection for this model were determine using the on Mean Decrease in Gini Impurity (MDGI) as shown in Table 4.2. MDGI represents the average reduction in node impurity attributable to each feature across all splits and across all the trees ($B = 500$) in the forest [Breiman, 2001]. These reported MDGI values are normalised such that all feature importances sum to one, enabling direct comparison of relative contributions across features. The highest contributor to the BCBM tumor detection using random forest was *original shape Elongation* with an MDGI value of 0.025. This means that elongation contributed 2.5% of the total impurity reduction across the entire forest, representing the largest contribution among all predictors. Similarly, *original glcm InverseVariance* contributed 2.2% (MDGI $= 0.022$), while *original shape Sphericity* contributed 2.0% (MDGI $= 0.020$).

The random forest model seem to have improved performance compared to LASSO regression with AUC, Accuracy and F1 score of 82.7%, 76.4% and 78.6%, respectively as given by Table 4.5.

Table 4.2: Top 5 features selected by Random Forest

Feature	Importance
original shape Elongation	0.025
original glvm InverseVariance	0.022
original shape Sphericity	0.020
original glvm MCC	0.018
original glvm Imc2	0.017

4.2.4 Baseline XGBoost (BXGB)

This section summarises results of the baseline XGBoost model with hyperparameter selected through 10-fold cross-validation as explained in section 3.4 as alluded by [Chen and Guestrin, 2016]. Feature importance in this model was assessed using normalised gain, a value that quantifies the contribution of each feature to reducing the model's loss function across all splits and boosting iterations [Chen and Guestrin, 2016]. Using this normalised gain XGboost top five ranked features ranked are given in Table 4.3. It shows that *Original glcm DifferencePercentage* was the most important feature (gain = 0.03), followed by *original shape Elongation* (gain = 0.020), with the remaining three features showing similar contributions (gain ranging from 0.017 to 0.018). Table 4.5 shows that although, the

Table 4.3: Top 5 features selected by XGBoost

Feature	Importance
original glvm DifferencePercentage	0.030
original shape Elongation	0.020
original shape LeastAxisLength	0.018
original firstorder Variance	0.018
original firstorder Skewness	0.017

AUC (84.2%) for XGboost is slightly above by 1.46% and 9.4% compared to that of random forest and LASSO regression, respectively. It's accuracy and F1 score are relatively than that of random forest by 2.84% and 3.86, respectively.

Baseline SVM (BSVM)

The SVM model was trained using all neuroimaging features with a radial basis function (RBF) kernel. The tuning parameters, cost (C) and kernel width (γ), were selected using 10-fold cross-validation as outlined in Section 3.6 [Cortes and Vapnik, 1995, Schölkopf and Smola, 2002]. SVM with a non-linear kernel does not provide inherent feature importance measures, therefore, permutation importance was used to assess the contribution of each feature. This method measures the decrease in model performance (AUC-ROC) when the values of a feature are randomly permuted while all other features are held constant [Altmann et al., 2010]. The permutation importance values of the top five features are reported in Table 4.4. It shows that feature *original shape Flatness* had the highest permutation importance (PI = 10.81), an indicator that shuffling its values can result in an average decrease of 10.81 percentage points in AUC. Additionally, *original glszm LowGrayLevelZoneEmphasis* (PI = 2.42) had a smaller effect on AUC, indicating a weaker contribution to model performance. Although slightly better than BLASSO

Table 4.4: Top 5 features selected by SVM

Feature	Importance
original_shape_Flatness	10.81
original_shape_Elongation	7.520
original_shape_Sphericity	4.530
original_shape_SurfaceVolumeRatio	2.994
original_glszm_LowGrayLevelZoneEmphasis	2.420

(4.63% higher), the BSVM model had a AUC 3.31% and 4.77% lower than BRNF and BSVM, respectively. The same trend is noticed fro accuracy and F1 score as shown in Table 4.5.

Table 4.5: Baseline Classification Models Performance

Model	Accuracy	Precision	Recall	F1 Score	Specificity	AUC
BLASSO	0.6651	0.6400	0.7547	0.6926	0.5755	0.7480
BRNF	0.7642	0.7187	0.8679	0.7863	0.6604	0.8274
BXGB	0.7358	0.7155	0.7830	0.7477	0.6887	0.8420
BSVM	0.7264	0.6967	0.8019	0.7456	0.6509	0.7943

4.2.5 LASSO Features Selected Classification Models (LASSO ML Models)

This selection present results Classification results for machine learning results with only 29 predictors obtained in section 4.2.2. These 29 LASSO-selected neuroimaging features were used to train the same four classification models being LASSO logistic regression, support vector machine , random forest, and XGBoost. All models were evaluated on a held-out test set of 212 samples (20% of the balanced dataset), with performance assessed using accuracy, precision, recall (Sensitivity), F1 Score, specificity, and AUC. The variable importance criteria was the same used in the baseline models. The results for variable importance when only LASSO selected variables are used for the different classifiers are shown in table 4.6 (a-d) are for logistic regression (LLASO), Random forest (LRNF), XGBoost (LXGB) and SVM (LSVM), respectively.

LASSO selected Logistic Regression Feature Importance(LLASSO)

LLASSO model identified a reduced subset of features with more stable coefficient magnitudes compared to BLASSO. Whilst the BLASSO model predictor in section 4.2.2 was dominated by first-order intensity features with very high coefficients

Table 4.6: LASSO Selected Top 5 Features for Different Classifiers

(a) LLASSO	
Feature	**Coefficient**
original_shape_SurfaceVolumeRatio	0.975
original_shape_Flatness	0.962
original_shape_Elongation	-0.498
original_shape_Sphericity	-0.487
original_glcm_JointEntropy	0.435
(b) LRNF	
Feature	**Importance Score**
original_shape_Elongation	100.0
original_shape_Sphericity	62.19
original_glcm_Imc1	49.40
original_firstorder_10Percentile	40.09
original_firstorder_Median	36.60
(c) LSVM	
Feature	**Permutation Importance**
original_shape_Elongation	0.076
original_shape_Flatness	0.071
original_shape_SurfaceVolumeRatio	0.028
original_glszm_LowGrayLevelZoneEmphasis	0.024
original_gldm_DependenceVariance	0.024
(d) LXGB	
Feature	**Gain**
original_shape_Elongation	0.082
original_glcm_Imc1	0.050
original_shape_Sphericity	0.050
original_shape_Flatness	0.048
original_shape_MinorAxisLength	0.046

rankings mostly likely due to multicollinearity in the high-dimensional feature space [Friedman et al., 2010, Ruppert, 2004], the LLASSO model that utilised only LASSO selected variables had smaller coefficients as shown in Table 4.6 (a). The number of significant predictor from a double LASSO reduced further from 29 to 26 features giving an overall reduction space by 23.4%. Moreover, predictors shift from intensity based to shape based ones including `original_shape_SurfaceVolumeRatio` ($\beta = 0.975$) and `original_shape_Flatness` ($\beta = 0.962$) which both indicates positive associations with likelihood of BCBM tumor. Meanwhile `original_shape_Elongation` ($\beta = -0.498$) and `original_shape_Sphericity` ($\beta = -0.487$) had weaker negative associations.

LASSO Selected Random forest (LRNF)

The LASSO selected random forest model (LRNF) just like baseline model BRNF was dominated by texture based features. A common feature for cancer related images. However, their GINI importance significantly increased by 97% and over 50% for elongation and sphericity as shown Table 4.6 (b). The LRNF model had the highest ranked of almost 100% compared to only 2.5% from the BLASSO model in Table 4.2.

LASSO Selected SVM (LSVM)

The SVM model showed consistent ranking on predictor variables through permutation importance consisting of shape features across both BSVM and LSVM as shown in Table 4.6 (c). This consistency ranking indicates stability of the model under a non-linear decision boundary. The LSVM model however, slightly reversed the order with Elongation (0.076) exceeding Flatness (0.071). Moreover, these values were slightly reduced from the BSVM values of (10.810) and 7.52 for flatness and elongation respectively. The difference in absolute importance values reflected changes in baseline model performance rather than changes in relative feature influence Breiman [2001]. Additional features such as texture variance features like `LowGrayLevelZoneEmphasis` and `gldm_DependenceVariance` also appeared in the LASSO reduced model.

XGBoost Feature Importance

Result in Table 4.6 (d) show XGBoost feature importance showed when only LASSO selected featured are used. It shows that LXGB model most important predictor is elongation (gain (0.082)), followed by `glcm_Imc1` (0.050) and sphericity (0.050) compared to the BXGB model which had much lowe gain values with the highest of gain = 0.030 (Table 4.4) from `original_glcm_DifferencePercentage`.

4.2.6 Performance of LASSO selected Neuroimaging Feature Models

The performance of classifier models for LASSO selected feature is given by Table 4.7. In general it shows that the is meaningful classification performance improvement across all classifiers and all metric when only LASSO selected predictors are used compared to baseline models in Table 4.5. In particular, the XGBoost classifier LXGB achieved the strongest overall performance, with an accuracy of 0.82, F1 score of 0.83, and AUC of 0.88 whilst Logistic regression still has the lowest performance. These correspond to performance improvements of 8.50%, 8.27%, and 4.25% for accuracy, F1 score, and AUC, respectively compared to the baseline model (BXGB). Similar improvements were noted for the SVM which demonstrated an increase of almost 7%, and 12% for AUC and specificity, respectively indicating a huge reduction in false positive classifications. The most consistent improvement across models was observed in specificity recording 16.9%, 14.4, 12.4%, 10.5% increase for LASSO regression, random forest, SVM, XGBoost classifiers, respectively. This pattern indicates that removing uninformative and redundant features reduced false positive classifications.

Table 4.7: Classification performance of Models on LASSO-selected neuroimaging features

Model	Accuracy	Precision	Recall	F1 Score	Specificity	AUC
LLASSO	0.7217	0.7684	0.6636	0.7122	0.7843	0.7971
LSVM	0.7925	0.7946	0.8091	0.8018	0.7745	0.8648
LRNF	0.7877	0.8095	0.7727	0.7907	0.8039	0.8587
LXGB	0.8208	0.8158	0.8455	0.8304	0.7941	0.8845

The improvement in the classifiers for LASSO Selected Feature models is further shown by the ROC curves for all models as given by Figure 4.3. It is clear that there exist strong discriminative ability across non-linear models, with all ensemble methods AUC exceeding an AUC of 0.85.

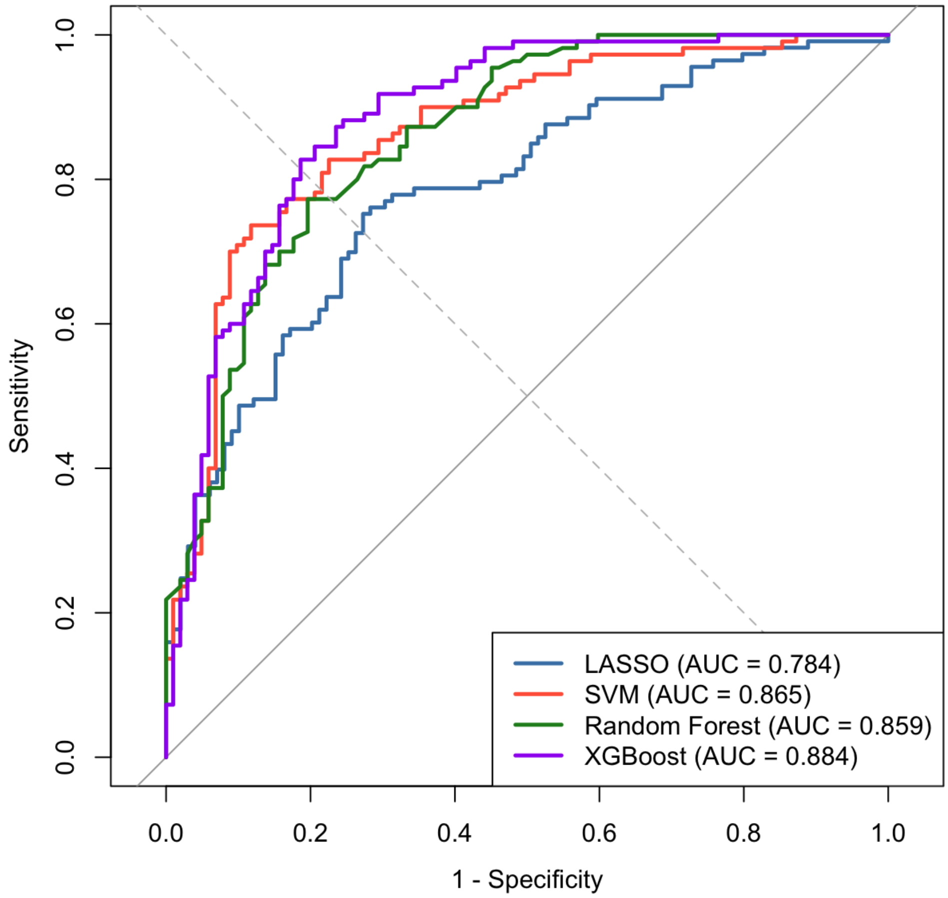

Figure 4.3: ROC curves for all models LASSO-selected features.

All models demonstrate good separation between classes, with curves positioned well above the diagonal reference line (AUC = 0.50). This indicated performance superior to random classification. Thus by introducing a feature selection approach through a LASSO regularisation

4.3 Identifying Differentially Expressed Genes

To answer objective 2, differential gene expression analysis was performed using LIMMA to identify genes whose expression levels differ significantly between primary breast cancer and BCBM. The hypothesis tested by LIMMA is that:
H_0: the mean log-expression level is equal between the primary breast cancer group and the BCBM group for any given probeset
H_1:the mean log-expression level for the primary breast cancer group differs from the BCBM group.
Using data in section 3.1.1 with a total 22,277 probesets on the Affymetrix U133A 2.0 array, a linear model was fitted and empirical Bayes shrinkage was applied to stabilise variance estimates across genes. This improved statistical power in the small paired dataset of 32 samples (16 primary breast cancer, 16 BCBM) [Smyth et al., 2003]. Our result shows that Of the 22,277 probesets tested, 105 had statistically significant differential expression between primary breast cancer and BCBM after Benjamini-Hochberg FDR correction at $P < 0.05$. These probest were then used o identity the unique genes they are associated with using the Database for Annotation, Visualization, and Integrated Discovery(DAVID) platform [Dennis Jr et al., 2003]. These 105 significant probesets resolved to 79 unique genes. It is important to note that the same gene can be represented by more than one probeset on the array. The LIMMA analysis was therefore conducted at the probeset level, and gene-level uniqueness was determined post hoc via DAVID. The top ten most significant probesets (ranked by p-value) are given by Table 4.8. The table shows that for each probesetID there is the corresponding gene name given as a gene symbol, the $\log_2$ fold change (logFC), which quantifies the direction and magnitude of expression change in BCBM relative to primary breast cancer, the average $\log_2$ expression across all 32 samples (AveExpr), which reflects the overall abundance of the transcript, and the Benjamini-Hochberg FDR-adjusted p-value (adj.P.Val), where values below 0.05 indicate statistically significant differential expression after correction for multiple testing. A negative logFC indicates that a gene is *downregulated* in BCBM compared to primary breast cancer that is, its mRNA abundance is lower in brain metastasis tissue than in the original tumor. A positive logFC would indicate upregulation. All ten probesets in Table 4.8 display negative logFC values, confirming a consistent pattern of gene suppression during the metastatic transition from primary breast tumor to brain. In this analysis, the same gene can appear in the table more than once under different probeset IDs. For example, *COL1A1* is represented by two probesets: probeset 217430_x_at, with a logFC of -3.01 and AveExpr of 11.88, and probeset 202311_s_at, with a logFC of -3.02 and AveExpr of 11.56. The closely concordant logFC values across these two probesets both indicating approximately 11-fold downregulation in BCBM provide converging evidence of *COL1A1* downregulation and increase confidence in this finding. The AveExpr of approximately 11-12 for both *COL1A1* probesets indicates that

this is a highly expressed gene overall, meaning the downregulation occurs from a high baseline and is therefore likely to have a substantive biological impact. As a general convention, logFC values with $|\text{logFC}| > 2$ are considered biologically meaningful, corresponding to at least a 4-fold expression change; values between 1 and 2 are considered moderate; and values below 1 are considered small, though all 105 probesets in this analysis met the FDR significance threshold regardless of fold change magnitude. *DPT*, the most significantly differentially expressed probeset (adj.P.Val $= 1.0 \times 10^{-4}$), shows a logFC of -5.23, corresponding to approximately 37-fold downregulation in BCBM, with an AveExpr of 8.07 indicating moderate overall expression. In contrast, *SPARC* shows a logFC of -2.17, corresponding to approximately 4-fold downregulation from an AveExpr of 12.83, indicating a highly expressed gene that undergoes a smaller but still statistically significant reduction in BCBM.

Table 4.8: Top 10 Differentially Expressed Probesets Ranked by Adjusted p-value

Affymetrix probeset IDs	Gene	logFC	AveExpr	adj.P.Val
213068_at	DPT	-5.23	8.07	1.0×10^{-4}
217430_x_at	COL1A1	-3.01	11.88	1.0×10^{-4}
213994_s_at	SPON1	-2.84	9.13	1.0×10^{-4}
206227_at	CILP	-4.20	8.12	1.8×10^{-4}
218469_at	GREM1	-3.85	8.04	2.0×10^{-4}
203477_at	COL15A1	-3.03	9.11	2.0×10^{-4}
201792_at	AEBP1	-3.36	12.65	2.2×10^{-4}
200665_s_at	SPARC	-2.17	12.83	2.4×10^{-4}
204457_s_at	GAS1	-3.71	7.97	3.1×10^{-4}
202311_s_at	COL1A1	-3.02	11.56	3.4×10^{-4}

This down gene regulation trend is further evidenced by Figure 4.4. The Figure presents boxplots of expression levels for the ten most significant genes on a $\log_{10}$ scale. It is clear that across all genes, the BCBM group (red) consistently shows lower expression than the primary breast cancer group (blue), visually confirming the uniform pattern of downregulation identified by LIMMA. For example *CTSK*, it is clear that the primary group shows a median of approximately 8-9 ($\log_{10}$), while the BCBM group has a substantially lower median of approximately 4-5 ($\log_{10}$), representing several orders of magnitude difference in expression. This pattern across all ten genes corroborates the negative logFC values in Table 4.8 and supports the conclusion that these genes are systematically downregulated during the metastatic transition from primary breast tumor to brain.

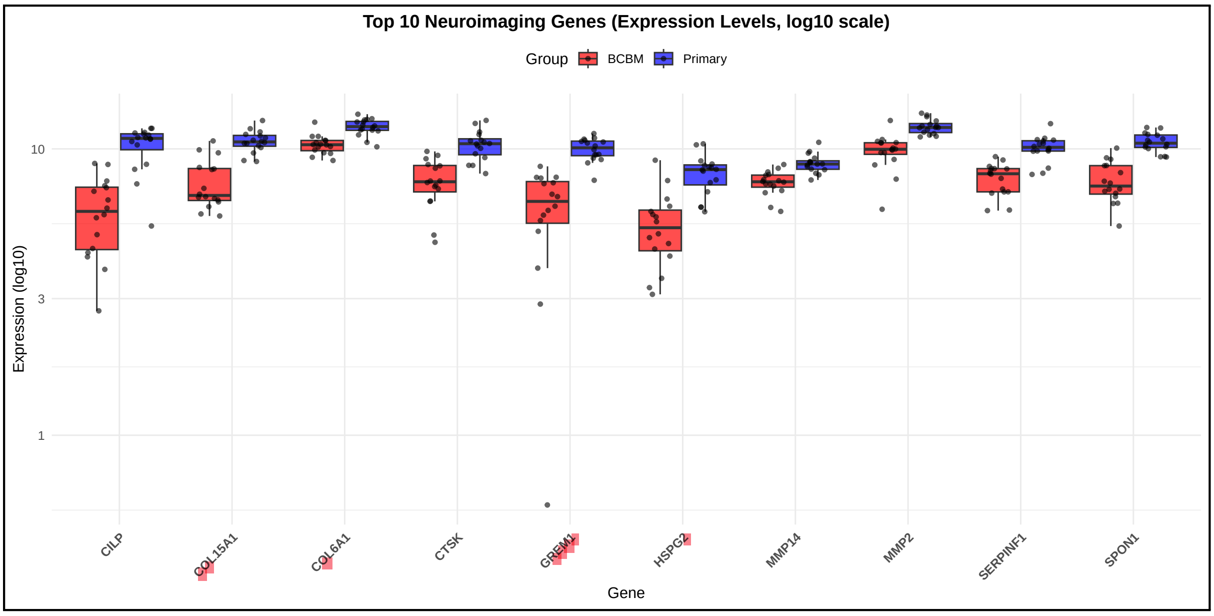

Figure 4.4: Expression levels of top 10 differentially expressed probesets

4.3.1 Identifying the Major Brain Regions Associated with BCBM

To identify the brain regions most likely to be structurally and functionally affected by BCBM, a correlation analysis between the identified DEGs and neuroimaging-derived phenotypes (NIDPs) was conducted using the `NeuroimaGene` R package [Bledsoe and Gamazon, 2024]. The package used reference data from the UK Biobank [Bycroft et al., 2018]. Our results indicates that only 17 genes out Of the 79 unique genes identified from the LIMMA analysis were available in the UK Biobank. This study therefore focused on the 17 identified genes from the UK-bibiobank. These genes included: SDHB, VCAN, LINC01140, GREM1, CTSK, SPON1, COL15A1, LCAT, COL6A2, SERPINF1, LRRC15, F2RL2, COL6A1, MMP14, HSPG2, MMP2, and CILP. Furthermore, the analysis integrated multiple imaging modalities, including T1-weighted MRI (for structural volume and cortical thickness), T2-weighted MRI, diffusion MRI (for white matter tract integrity), and resting-state fMRI (for BOLD signal). Brain regions were defined using both the Desikan–Killiany and Destrieux atlases. Analysing each modality or atlas individually did not yield significant associations due to the small number of cases per stratum. Significant gene-NIDP associations were only identified when all modalities and atlases were combined.

4.3.2 Identifying the Brain Regions

Across the 17 genes and all modalities combined, significant associations were identified spanning a total of 33 distinct NIDPs. Figure 4.5(a) ranks the 17 genes by the number of significant NIDP associations. CTSK had the greatest number of associations (approximately 15 NIDPs), followed by SDHB (10 NIDPs) and VCAN (7 NIDPs). Several genes including LINC01140, COL6A1, COL6A2, LRRC15, F2RL2, LCAT, COL15A1, and HSPG2 had few associations (1-3 NIDPs each), while SERPINF1, MMP14, and LCAT each had only a single significant NIDP association. Figure 4.5(b) groups these associations by anatomical brain region and shows that majority of associations fell under a broad "Other" grouping encompassing white matter tracts and miscellaneous cortical parcels. Among the four anatomically defined ROIs, the cerebellum showed the greatest number of gene-NIDP associations, followed by the thalamus, the cingulate cortex, and the frontal lobe.

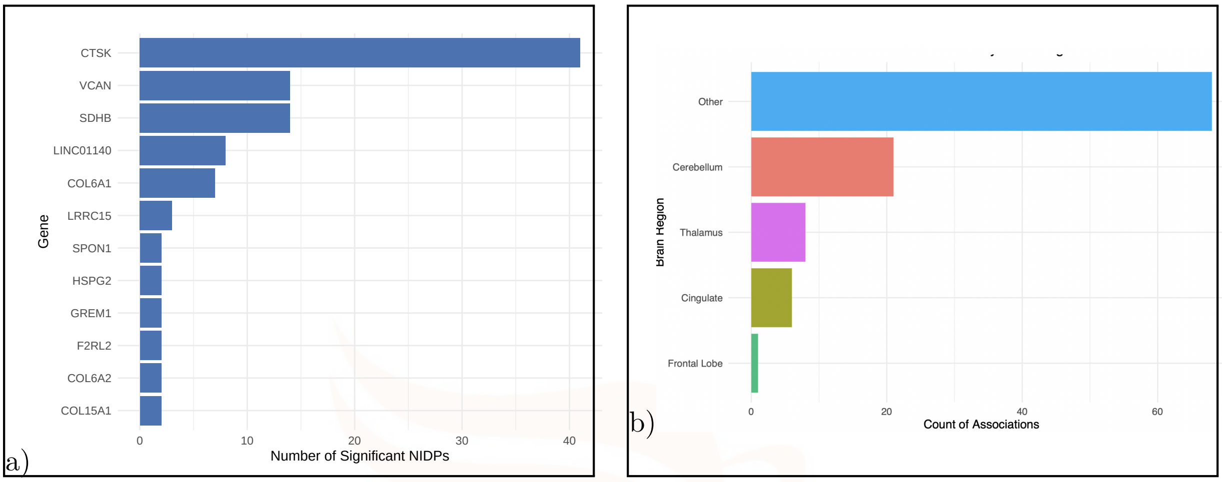

Figure 4.5: Top genes by number of NIDP associations and Gene-NIDP associations by brain region.

Summary of Gene-NIDP Associations

Table 4.9 provides the full summary of all significant gene-NIDP associations. Each row reports the gene, the specific NIDP, and the direction of association (positive or negative). A positive association indicates that higher genetically regulated expression of a gene is linked to greater volume, thickness, intensity, or connectivity in the associated brain region while a negative association indicates the inverse [Boire et al., 2020]. For example, CTSK shows exclusively positive associations across 15 NIDPs spanning the thalamus (bilateral intensity of the thalamus-proper), frontal orbital cortex, cingulate gyrus, and several white matter tracts including the striatum and uncinate fasciculus. This indicates that higher CTSK expression is linked to greater structural and white matter integrity in these regions, regions whose involvement is consistent with sensory, cognitive, and motor functions known to be disrupted in BCBM [Boire et al., 2020]. VCAN shows exclusively positive associations with seven cerebellar NIDPs across bilateral cerebellar lobules, linking higher VCAN expression to greater cerebellar volume. This is consistent with the known role of the cerebellum in motor coordination deficits observed in patients with BCBM [Carvalho et al., 2025]. SDHB shows positive associations predominantly in frontal white matter pathways (external capsule, fornix/stria terminalis) and frontal cortical parcels (pars orbitalis, pars triangularis). Genes such as MMP2

and MMP14 showed negative associations with cingulate and anterior corona radiata tracts, suggesting that downregulation of these matrix metalloproteinases in BCBM may correspond to white matter disruption in these regions [Rostami et al., 2016].

Table 4.9: Summary of significant DEG-NIDP associations

Gene	**Phenotype (NIDP)**	**Effect**
CILP	aparc-pial_lh_area_posteriorcingulate	Positive
COL15A1	rfMRI_amplitudes_(ICA100_node_1), aparc-pial_rh_area_entorhinal	Positive, Negative
COL6A1	aparc-a2009s_lh_thickness_G-temp-sup-Lateral, rfMRI_(ICA100_edge_1243), rfMRI_(ICA100_edge_629)	Positive, Negative
COL6A2	rfMRI_(ICA100_edge_1273), IDP_T1_FAST_ROIs_R_cerebellum_crus_II	Positive
CTSK	aseg_lh_intensity_Thalamus-Proper, aseg_lh_intensity_Accumbens-area, aseg_rh_intensity_Thalamus-Proper, IDP_dMRI_ProbtrackX_OD_str_l, IDP_T1_FAST_ROIs_R_front_orb_cortex, IDP_dMRI_TBSS_ISOVF_Cingulum_cingulate_gyrus_R, IDP_dMRI_ProbtrackX_FA_str_l, IDP_dMRI_ProbtrackX_ISOVF_unc_l, IDP_T1_FAST_ROIs_R_sup_temp_gyrus_post, IDP_dMRI_ProbtrackX_OD_str_r, IDP_dMRI_ProbtrackX_FA_str_r, aparc-DKTatlas_lh_area_medialorbitofrontal, aparc-a2009s_lh_area_S-temporal-sup, IDP_dMRI_ProbtrackX_MO_str_l, IDP_dMRI_ProbtrackX_ISOVF_slf_r	Positive
F2RL2	rfMRI_(ICA100_edge_800)	Positive
GREM1	rfMRI_(ICA100_edge_1402), aparc-a2009s_lh_thickness_G-temporal-middle	Negative, Positive
HSPG2	aparc-a2009s_rh_volume_G-front-inf-Orbital, aparc-Desikan_rh_volume_parsorbitalis	Negative
LCAT	rfMRI_(ICA100_edge_383)	Negative
LINC01140	IDP_dMRI_TBSS_OD_Superior_cerebellar_peduncle_L, HippSubfield_lh_volume_molecular-layer-HP-body, IDP_dMRI_TBSS_L3_Sagittal_stratum_R, IDP_dMRI_TBSS_MO_Superior_cerebellar_peduncle_L, HippSubfield_lh_volume_Whole-hippocampal-body	Negative
LRRC15	aparc-a2009s_lh_area_S-temporal-sup, aparc-a2009s_lh_area_G+S-subcentral	Negative
MMP14	IDP_dMRI_TBSS_OD_Anterior_corona_radiata_L	Negative
MMP2	IDP_dMRI_TBSS_L1_Cingulum_hippocampus_R	Negative
SDHB	IDP_dMRI_TBSS_OD_External_capsule_R, aseg_global_volume_CC-Posterior, IDP_dMRI_TBSS_OD_External_capsule_L, IDP_dMRI_TBSS_MO_Fornix_cres+Stria_terminalis_R, IDP_dMRI_TBSS_L1_External_capsule_R, aparc-Desikan_rh_volume_parstriangularis, aparc-DKTatlas_lh_volume_parsorbitalis, aparc-a2009s_rh_volume_G-front-inf-Triangul, aparc-DKTatlas_rh_volume_parstriangularis, aparc-pial_rh_area_parstriangularis	Positive
SERPINF1	rfMRI_(ICA100_edge_823)	Negative
SPON1	IDP_dMRI_TBSS_L3_Inferior_cerebellar_peduncle_L	Negative
VCAN	IDP_T1_FAST_ROIs_L_cerebellum_VI, IDP_T1_FAST_ROIs_L_cerebellum_V, IDP_T1_FAST_ROIs_L_cerebellum_I-IV, IDP_T1_FAST_ROIs_R_cerebellum_VI, IDP_T1_FAST_ROIs_R_cerebellum_crus_I, IDP_T1_FAST_ROIs_R_cerebellum_VIIb, aseg_lh_volume_Cerebellum-Cortex	Positive

4.3.3 Variable Selection by Genomic Correlations

Based on the NeuroimaGene analysis, the significant gene–NIDP associations in Table 4.9 were used to select neuroimaging features for the classification models. Only features mapped to four brain regions strongly associated with BCBM-relevant gene expression were retained. The regions were the cerebellum, the thalamus, the cingulate cortex, and the frontal lobe. These regions accounted for the majority of anatomical associations in Figure 4.5(b) and have established neurological significance in BCBM, where lesions are known to impair coordination, sensory processing, cognition, and mood regulation, respectively [Boire et al., 2020, Gallivanone, 2022]. Applying this filter to the full neuroimaging dataset, retained all 107 neuroimaging features across 982 observations from 132 patients with 304 BCBM cases and 678 primary breast cancer cases. This approach, referred to as *genomically-identified variable selection*, provided a biologically informed, dimensionality-reduced input to the classification models.

4.4 Classification models using Genom-identified Variables

Genom Selected LASSO Logistic Regression (GLASSO)

Table 4.10(a), shows the top 5 predictors from logistic regression classifier with a LASSO for BCBM tumor when the predictor feature space restricted to genomically relevant regions. Unlike the BLASSO and LLASSO which had intensity and shape features dominating,respectively, with positive associations. The genom associated classification had both positive correlations from the texture and intensity features dominated the top predictors with `original_glszm_HighGrayLevelZoneEmphasis` ($\beta = 3.73$), `original_glcm_Autocorrelation` ($\beta = 2.98$), and `original_glcm_JointEntropy` ($\beta = 2.075$) ranking as the most, with no shape feature appearing among the top five.

Genomic Random forest (GRNF)

The random forest classier feature importance rankings showed a high degree of consistency across all models in baseline, Laso selected and DEG-associated feature selection analysis. As shown in Table 4.10(b) and BRNF and LRNF models shpericity and elongation features are ranked among the top predictors for BCBM tumor. Sphericity maintained the highest gini entropy gain of 21.297 when feature selection is via genomic signature associations. Slight differences where however, observed for GRNF as it includes first-order statistics features as major predictors including skewness (13.40), Median(13.74) as opposed to graylevel features in the LRNF model.

Genomic Selected XGBoost Classifier (GXGB)

The top 5 features from the GXGB model represents the major predictor for BCBM tumor when only features from BCBM genom informed regions are utilised as presented in Table 4.10(c). Again elongation (14.44) and sphericity (11.13) are among the major predictors although LowGLE tops the list with gain of 16.45. Additional differences were observed in the presence of texture features such as `original_glcm_Correlation`, which ranked fourth in the DEG analysis (12.35) but was not among the top-ranked features in the LASSO based selection.

Genomic Selected SVM (GSVM)

SVM permutation importance rankings demonstrated the highest level of consistency across the two analysis. As shown in Table 4.10(d), Elongation, Flatness and Sphericity remain top predictors for BCBM when using SVM classifier just like LSVM and BSVM achieving a permutation importance of 3.17, 2.41 and 2.30, respectively. However, GSVM additionally had gray level feature like `original_gldm_DependenceNonUniformityNormalized` (2.348), which were not among the top predictors in the LSVM model.

4.4.1 Performance of Genomically Identified Neuroimaging Features

Table 4.11 shows how the four models performed when trained using the genomically identified neuroimaging features. All models performed better than random chance, with AUC values between 0.635 (GLASSO) and 0.774 (GRNF). However, these results are lower than those obtained using the full feature set (Table 4.5) and the LASSO-selected features (Table 4.7). Overall, the random forest model (GRNF) performed best. It had the highest accuracy (0.672), recall (0.786), F1 score (0.688), and AUC (0.774), meaning it identified more true BCBM cases than the other models. GSVM came next, with an AUC of 0.736 and fairly balanced performance across metrics. GXGB performed similarly to GSVM, with an AUC of 0.738 and F1 score of 0.667. GLASSO performed the weakest overall, with the lowest accuracy (0.615) and AUC (0.635). Specificity was fairly similar across all models, ranging from 0.576 to 0.591, meaning they performed almost the same in identifying non-metastatic cases. However, all models showed higher recall than specificity, meaning they were better at detecting BCBM cases than ruling them out.

Table 4.10: Genom Identified Top 5 Predictors across Classifiers

Feature	Importance
(a) GLASSO	
	Coefficient
original glszm HighGrayLevelZoneEmphasis	3.73
original glcm Autocorrelation	2.98
original glcm JointEntropy	2.07
original shape SurfaceArea	1.94
original glrlm ShortRunLowGrayLevelEmphasis	1.82
(b) GRNF	
	Gini Impurity
original shape Sphericity	21.30
original shape Elongation	19.36
original glcm Correlation	16.92
original firstorder Median	13.74
original firstorder Skewness	13.41
(c) GXGB	
	Gain entropy
original gldm LargeDependenceLowGLE	16.45
original shape Elongation	14.44
original firstorder 90Percentile	12.48
original glcm Correlation	12.35
original shape Sphericity	11.13
(d) GSVM	
	Permutation
original_shape_Elongation	3.17
original_shape_Flatness	2.41
original_gldm_DependenceNonUniformityNormalized	2.35
original_shape_Sphericity	2.30
original_glszm_ZonePercentage	1.95

4.4.2 Performance of DEG-Associated Neuroimaging Feature Models

Table 4.11: Model performance using original radiomic features

Model	Accuracy	Precision	Recall	F1-score	Specificity	AUC
GLASSO	0.6148	0.5714	0.6429	0.6050	0.5909	0.6347
GSVM	0.6475	0.5970	0.7143	0.6504	0.5909	0.7362
GRNF	0.6721	0.6111	0.7857	0.6875	0.5758	0.7739
GXGB	0.6557	0.6000	0.7500	0.6667	0.5758	0.7376

Feature Importance Across Models

Figure 4.6 compares feature importance scores from GRNF and GXGB model trained on the DEG-associated feature set. It is clear that features such as elongation, sphericity, inverse variance, and GLCM-based metrics consistently rank among the most important predictors across both models. Thus BCBM tumor models can be dtermined accurately when these features have a contribution as they may determine iscriminative accuracy. Although important these feature as determined by tree based importance do not imply causal relationships, as tree-based importance metrics may be biased towards features with higher variability or more potential split points. Therefore it is alsways ecommended to interpret them alongside more robust interpretability techniques such as SHAP [Breiman, 2001].

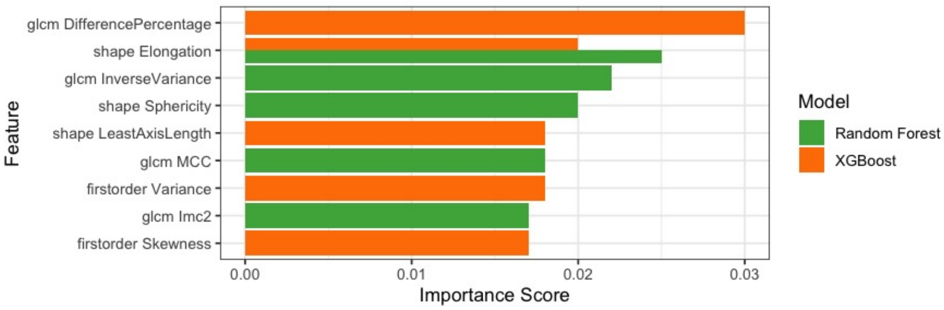

Figure 4.6: Feature importance scores for Random forest and XGBoost

SHAP-Based Global Feature Importance

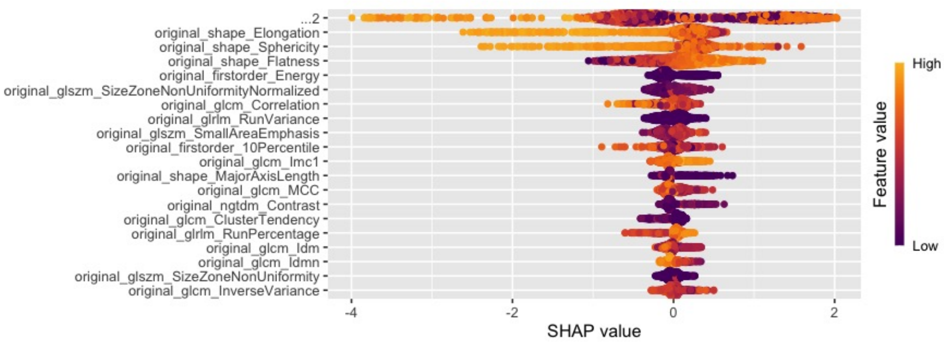

Figure 4.7: Mean absolute SHAP values for XGBoost model

To address the limitations of traditional feature importance measures, SHAP (SHapley Additive exPlanations) values were used to quantify feature contributions. Figure 4.7 shows the mean absolute SHAP values, representing the average impact of each feature on model predictions. Unlike traditional importance scores, SHAP values are grounded in cooperative game theory and provide a consistent and locally accurate decomposition of predictions into individual feature contributions [Fryer et al., 2021, Rozemberczki et al., 2022]. The results indicate that shape-based features, particularly elongation and sphericity, have the largest influence on classification outcomes, further supporting the conclusion that tumor morphology plays a central role in distinguishing metastatic from non-metastatic cases within the genomically-identified brain regions.

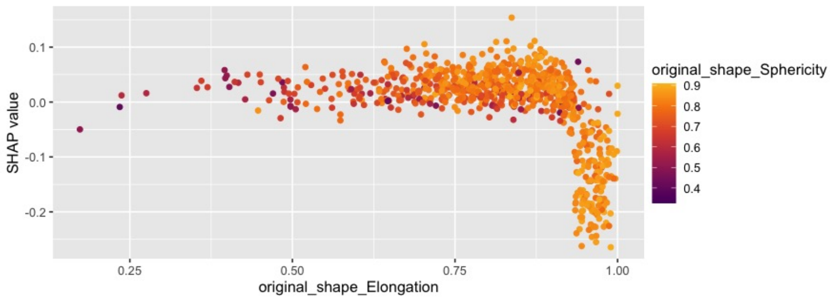

Figure 4.8: SHAP dependence plot for elongation

Figure 4.8 illustrates the relationship between elongation and its SHAP contribution. Higher elongation values are associated with increased SHAP values, indicating a higher probability of tumor classification.

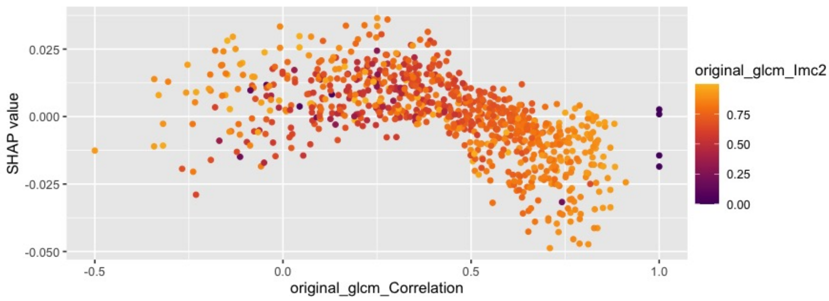

Figure 4.9: SHAP dependence plot for GLCM correlation

Figure 4.9 shows the effect of GLCM correlation on model predictions. The relationship appears non-linear and shows interaction effects with other features, further supporting the need for flexible models capable of capturing complex dependencies.

Calibration Analysis

Figure 4.10 presents the calibration curves comparing predicted probabilities with observed outcomes for each model. Ideally, well-calibrated models should align

closely with the diagonal reference line, indicating agreement between predicted and observed event probabilities. Overall, the calibration curves indicate agreement between predicted and observed probabilities across the models. This is consistent with the strong discriminative performance observed in Table 4.7, where all models achieved AUC values above 0.70, with XGBoost demonstrating the highest discrimination (AUC = 0.815).

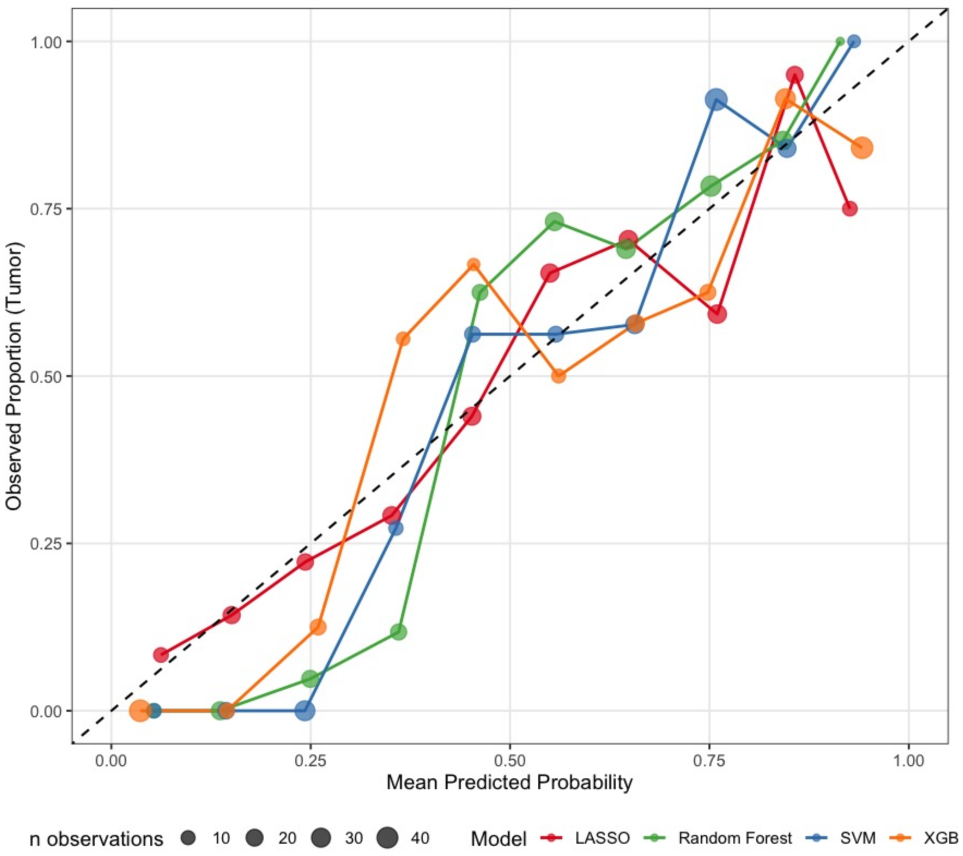

Figure 4.10: Calibration curves for classification models

Across all models used in the baseline, LASSO selected and genom idendified selection methodds, the XGBoost and random forest were overally good performers with the least being logistic regression. The LASSO feature selection model however proved to be a better method compared to genom identified selection as its models as shown by LLASSO, LRNF, LSVM, LXGB outperformed both the baseline models (BLASSO, BRNF, BXGB, BSVM) and Genom identified models (GLASSO, GRNF, GXGB, GSVM) models. Moreover, across all models, shape-based neuroimaging features particularly elongation, sphericity, and flatness were shown to be the most consistent and robust predictors of BCBM status. In particular, the inclusion of DEG-associated features introduced additional texture and intensity-based signals, especially in LASSO and XGBoost models. The consistency of findings across both linear and non-linear models suggests that tumor shape features represent fundamental imaging features associated with brain metastasis. In addition, the contribution of texture related features in DEG-informed models supported that integrating genomic information improves the biological interpretability of neuroimaging features.

4.5 Discussion

This chapter presented the results of neuroimaging-based classification of BCBM. LASSO-based dimensionality reduction consistently improved model performance compared to the full feature set. LASSO and DEG-guided feature selection identified complementary neuroimaging features with shape features dominating in LASSO and texture and intensity features highlighted in DEG-guided selection. Overall, the results show that a reduced and biologically informed set can achieve strong classification performance while improving model interpretability.

Across all models used in the baseline, LASSO selected and genom idendified selection methodds, the XGBoost and random forest were overally good performers with the least being logistic regression. The LASSO feature selection model however proved to be a better method compared to genom identified selection as its models as shown by LLASSO, LRNF, LSVM, LXGB outperformed both the baseline models (BLASSO, BRNF, BXGB, BSVM) and Genom identified models (GLASSO, GRNF, GXGB, GSVM) models. Moreover, across all models, shape-based neuroimaging features particularly elongation, sphericity, and flatness were shown to be the most consistent and robust predictors of BCBM status. In particular, the inclusion of DEG-associated features introduced additional texture and intensity-based signals, especially in LASSO and XGBoost models. The consistency of findings across both linear and non-linear models suggests that tumor shape features represent fundamental imaging features associated with brain metastasis. In addition, the contribution of texture related features in DEG-informed models supported that integrating genomic information improves the biological interpretability of neuroimaging features.

5. Discussion

This chapter discusses the findings of this study organised by variable selection guided by genomic data, variable selection using LASSO regularisation, a comparative analysis of both selection, and the classification performance of the machine learning methods evaluated.

5.1 Variable Selection Using LASSO

5.1.1 LASSO as a Dimensionality Reduction

LASSO-based variable selection reduced the neuroimaging feature space from 107 to 29 features. This was a dimensionality reduction of 24.3% . This result is consistent with findings reported in neuroimaging studies, where LASSO regularisation has been shown to reduce feature dimensionality by 20–60% while preserving or improving classification performance Aerts et al. [2014], Parmar et al. [2015]. In the full model, coefficients ranged from -21.506 to 19.545, while in the reduced model this range decreased to -0.498 to 0.975. This is consistent with the expectation that L1 regularisation reduces inflated coefficients caused by multicollinearity Ruppert [2004], Tibshirani [1996]. There was also a shift in the type of important features. The full model was mainly driven by intensity-based features (`RootMeanSquared`, `Mean`). After feature selection, shape-based features (`SurfaceVolumeRatio`, `Flatness`) became more important. This suggested that the large coefficients in the full model could have been influenced by high correlation among intensity features, rather than representing true biological effects [Ruppert, 2004].

5.1.2 Shape Features as Dominant LASSO-Selected

The dominance of shape-based features after LASSO selection is shown in Table ??. Several studies have shown that morphological features are reliable indicators of tumor aggressiveness and metastatic potential [Aerts et al., 2014, Gillies et al., 2016]. In a study by, Aerts et al. [2014] identified shape-related features, such as compactness and volume, as important predictors of clinical outcomes in lung and head-and-neck cancers. They linked these features to tumor growth patterns, where irregular shapes reflect more invasive behaviour [Aerts et al., 2014]. In this study, the positive LASSO coefficient for `original_shape_SurfaceVolumeRatio` ($\beta = 0.975$), observed in the reduced feature model, supports this interpretation. A higher surface-to-volume ratio indicates a more irregular tumor shape, which is often associated with invasive and metastatic growth in brain lesions Achrol et al. [2019]. Furthermore, the negative coefficient for `original_shape_Elongation` ($\beta = -0.498$), also reported in the LASSO-selected feature set, suggested that more elongated lesions are associated with lower odds of BCBM. This may reflect differences in tumor morphology between primary breast tumors and metastatic brain lesions. Similar observations have been reported by Kuo and Jamshidi [2014], who found that brain metastases tend to appear more spherical and well-defined, while primary tumors may show more elongated shapes. Importantly, these findings are supported by the consistency of shape features across multiple models. As shown in Table 4.5, features such as `original_shape_Elongation` and `original_shape_Sphericity` were also highly ranked in random forest, SVM, and XGBoost models, reinforcing their role as stable and robust predictors of BCBM across different modelling approaches.

5.2 Variable Selection Guided by Genomic Data

5.2.1 Genome-identified Feature Selection

The integration of gene expression data into neuroimaging feature selection provided a biologically informed approach to dimensionality reduction. Traditional neuroimaging analyses select features based only on associations with clinical outcomes, without considering the underlying biological mechanisms [Gillies et al., 2016, Lambin et al., 2012]. In this study, genome-identified variable selection using differentially expressed genes linked the neuroimaging features to molecular pathways known to be involved in BCBM. This improved the biological relevance and interpretability of the selected features [Aerts et al., 2014]. Furthermore,this approach is supported by radiogenomics, which suggests that imaging features reflect the underlying genomic profile of tumors Gillies et al. [2016]. For example, Kuo and Jamshidi [2014] showed that neuroimaging features of brain tumors are associated with gene expression patterns related to cell proliferation, angiogenesis, and extracellular matrix remodelling. Similarly, Zinn et al. [2011] found that texture features derived from GLCM are linked to genes involved in tumor invasiveness and hypoxia. These findings suggested that DEG-informed neuroimaging features capture biologically meaningful information that may not be identified through statistical selection alone.

5.2.2 Interpretation of DEG-Selected Features

The DEG-based feature selection identified texture and intensity features as the main predictors of BCBM. In particular, `original_glszm_HighGrayLevelZoneEmphasis`, `original_glcm_Autocorrelation`, and `original_glcm_JointEntropy` had the largest LASSO coefficients. The importance of these features is biologically informative. High gray-level zone emphasis measures the presence of large, high-intensity regions within the tumor [Breiman, 2001]. These regions may reflect high cellular density, necrosis, or increased vascular activity [Zwanenburg et al., 2020a]. This is consistent with known characteristics of BCBM, which often show dense tumor tissue, blood–brain barrier disruption, and surrounding oedema on MRI [Achrol et al., 2019]. The feature original_glcm_Autocorrelation was also highly important. This feature measures how similar neighbouring pixel intensities are. Higher values indicate more uniform texture patterns [Zwanenburg et al., 2020b]. In metastatic lesions, this may reflect more regular and well-defined tumor structure. Brain metastases often grow in a more uniform and clearly defined way compared to primary brain tumors [Achrol et al., 2019]. This is supported by the positive LASSO coefficient ($\beta = 2.981$), which shows that higher autocorrelation is associated with increased likelihood of BCBM. The correlation analysis identified 105 differentially expressed probesets between primary breast cancer and BCBM corresponding to 79 distinct genes and 17 of them had NIDP available from the UK Biobank. Four major brain regions namely thalamus, cerebellum, frontal lobes and the cingulate cortex were associated with genes *CTSK*, *VCAN*, and *SDHB* which stood out. These observations emphasized how radiogenomics can show what genes can be implicated in brain structure and function differences. Moreover, the use of multiple MRI modalities (T1, T2, diffusion, and functional MRI) have allowed both structural and functional associations to be detected. This was similar to previous studies that demonstrated that combining imaging and genomic information can show biological patterns [Fan et al., 2020, Guyon et al., 2002, Zwanenburg et al., 2020a].

5.2.3 Brain Region Associations and Neurological Significance

The thalamus was associated with the gene *CTSK*. A study showed that metastases in this region often result in sensory disturbances, including impaired tactile perception, difficulties in pain modulation, and coordination problems [Medicine, 2025]. Cerebellum was another important region, associated to *VCAN*. The National Human Genome Research Institute [2023] reported that lesions in this part of the brain are known to cause ataxia, balance problems, and deficits in fine motor control in up to 20% of affected patients. It is also supported by the gene–region association identified in our study. Associations, specifically with *SDHB* and *CTSK* were shown by the fronal lobe. Metastases in this region is known to contribute to cognitive and behavioural changes, including executive dysfunction and episodes of confusion which is also seen in our study[Boire et al., 2020]. Lastly, cingulate cortex was linked to *SDHB*, an observation in National Human Genome Research Institute [2023] which that reported that showing that disruptions in this region can lead to mood disorders, attention difficulties, and even autonomic irregularities. These results on the gene-brain region associations provided knowledge into how BCBM affects neurological function. Through the connection of DEG's with with structural and functional patterns in the brain. The study moved beyond offering a purely predictive model (Black-box) but instead provided an interpretable biological explanation of how metastatic processes can influence neurological function[Kallah-Dagadu et al., 2025]. This is imperative because small or slight changes in the regions affected mentioned above can have a significant effect on patient survival, treatment response and the quality of life.

5.2.4 DEG-Associated Neuroimaging Feature Models

The DEG-associated neuroimaging models helped reveal biologically meaningful features linked to tumor heterogeneity and morphology. Previous radiogenomics studies have shown that texture and shape features are strongly associated with tumor biology and can improve predictive models of metastasis [Demetriou et al., 2024, He et al., 2024]. Consistent with these findings, our analysis identified similar patterns. LASSO logistic regression primarily selected texture features, random forest, identified both shape and texture and XGBoost and SVM emphasized dependence-based texture features alongside shape. First-order intensity features also contributed, in line with observations from neuroimaging-only models presented in 4.2. Additionally, tree-based models gave higher importance to both shape and texture, suggesting that within these regions, internal structure and overall form together carry the strongest information. This is consistent with previous reports that ensemble models capture complex interactions among features better than linear models [Saha et al., 2018]. SVM models were mostly driven by shape. Using permutation importance for SVM, which measures how much model performance drops when a feature is shuffled as described in Altmann et al. [2010], we found that changes in tumor geometry had the greatest effect on classification. Overall, these results support a multi-feature view of BCBM, where no single measure is sufficient. Combining texture, intensity, and shape, particularly in biologically relevant regions such as the cerebellum and thalamus, provides the most reliable detection and the clearest clinical interpretation. Although DEG-associated neuroimaging feature models showed slightly lower overall performance compared to the full neuroimaging feature set, they achieved improved recall, especially in regions like the thalamus and cerebellum, which is critical for reducing false negatives in metastasis detection. These findings confirm that integrating genomics identifies predictive features and enhances biological interpretability, which supports that gene-linked brain imaging features can explain tumor behavior and heterogeneity [Li, 2022]

Tumor Detection Efficiency

A radiogenomic study by Orton et al. [2023] showed that using fewer, carefully selected features kept models easy to interpret while still performing well. Similarly, in head and neck cancer, combining neuroimaging with gene data gave better classification than using neuroimaging alone, even with a limited number of features [Spielvogel et al., 2023]. By restricting features to regions linked to the available DEGs, the model focused on features that are genomical tied to metastatic processes (texture and shape changes in specific ROIs) rather than broad and nonspecific features. In addition, because each positive decision is associated to named genes and brain regions, the resulting detections are also more interpretable, associating classes to plausible changes rather than meaningless predictors. Furthermore, this supported the use of neuroimaging features associated with DEG'S as sensitivity tools for metastasis detection, with clear links to underlying biology and practical ways to BCBM targeted treatment strategies.

5.3 A Comparative Analysis of LASSO-Based and Genom-identified Variable Selection

The comparison of the LASSO-selected and DEG-associated feature importance rankings shows both similarities and differences. These findings highlight the complementary nature of the two feature selection approaches. The most consistent result across both analyses is the importance of `original_shape_Elongation`. This feature ranked among the top features in the random forest, SVM, and XGBoost models under both frameworks. This consistency across methods and models suggests that lesion elongation is a stable and reliable neuroimaging biomarker for BCBM classification [Aerts et al., 2014, Zwanenburg and et al., 2020]. However, the two approaches differ in the secondary features they identify. The DEG-based analysis consistently highlighted texture and intensity features, especially those derived from GLCM and GLSZM. On the other hand, the LASSO-based analysis placed more emphasis on first-order intensity features and some texture measures. This difference can be explained by how each method selects features. LASSO selects variables based only on their statistical relationship with the outcome. In contrast, DEG-based selection incorporates biological information from gene expression data. These genes are known to influence tumor characteristics such as vascularity, cellularity, and tissue structure. These biological processes are reflected in imaging as variations in texture and intensity [Gillies et al., 2016, Kuo and Jamshidi, 2014, Zinn et al., 2011]. Although the models trained on DEG-associated features showed slightly lower overall performance compared to the LASSO-based models, they remained comparable across all evaluation metrics. For example, GSVM, GRNF, and GXGB achieved AUC values of 0.736, 0.774, and 0.738, respectively. This indicated that meaningful classification performance was retained even with a reduced, biologically informed feature set, supporting the stability, interpretability, and biological relevance of these features.

5.4 Classification Performance Across Machine Learning Methods

5.4.1 Performance on Full Neuroimaging Features

All four models trained on the full 107-feature dataset performed well above chance, with AUC values ranging from 0.748 (LASSO Logistic Regression) to 0.842 (XGBoost). These values are consistent with previous neuroimaging studies in brain metastasis classification, which typically report AUC values between 0.70 and 0.90 [Fan et al., 2020, Kuo and Jamshidi, 2014]. In general, non-linear models (random forest, SVM, XGBoost) outperformed the linear LASSO model. This pattern is expected in high-dimensional neuroimaging data, where relationships between features and outcome are often non-linear [Breiman, 2001, Chen and Guestrin, 2016, Parmar et al., 2015]. Across all models, specificity was lower than sensitivity (0.5755-0.6887). This indicated that some non-metastatic cases may been incorrectly classified as metastatic. This is a common issue in medical classification problems, especially when models are tuned to prioritise sensitivity due to clinical importance of avoiding missed metastases [Fawcett, 2006].

5.4.2 Performance Improvement Following LASSO Selection

After LASSO-based feature selection, performance improved across all models. The most consistent improvement was observed in specificity, which increased across models. XGBoost achieved the best overall performance after feature selection (Accuracy = 0.7642, AUC = 0.8241, Recall = 0.8491). Random forest also performed strongly, with the highest AUC in the full-feature model (0.8274) and stable performance after reduction. These results suggest that removing redundant and correlated features improves generalisation, particularly for ensemble methods [Chen and Guestrin, 2016, Ruppert, 2004]. The improvement after LASSO indicates that many of the original features were redundant or noisy, and that a more compact feature set improves model stability and interpretability. The AUC values obtained in this study (approximately 0.77 to 0.84) are consistent with previous work in neuroimaging-based brain metastasis classification. For example, Lin and Winer [2007] reported AUC values around 0.81-0.87, while Li [2022] reported similar performance ranges in neuroimaging classification tasks. This suggested that the models developed in this study perform within the expected range for this type of data. The repeated importance of `original_shape_Elongation` across models is also consistent with prior studies showing that shape irregularity is a strong indicator of tumor aggressiveness and metastatic potential [Parmar et al., 2015, Saha et al., 2018]. This supports the biological relevance of shape-based neuroimaging features in BCBM classification.

5.5 Contributions of the Study

This study makes several methodological and applied contributions to the analysis of BCBM using radiogenomic data. Firstly, it addresses the challenge of high dimensionality by using supervised dimensionality reduction using LASSO logistic regression improving model stability and feature selection. Secondly, it develops a genom-identified neuroimaging feature selection by integrating DEG's with NIDPs as outlined in Chapter 3 and 4. Correlation analysis enabled the identification of biologically relevant brain regions and associated imaging features. Furthermore, the study provided a comparative evaluation of machine learning models across different feature selection strategies. This then presented a reproducible radiogenomic modelling approach that integrates MRI-derived features and gene expression data using a publicly available dataset and established statistical learning techniques. Most importantly, it supported the development of clinically interpretable models and reduced dependence on purely "black-box" model outputs.

6. Conclusion

6.1 Conclusion

This study found that combining neuroimaging features with genomic data enabled machine learning models to classify BCBM with high perfomance. It also revealed associations between tumor imaging patterns, gene expression, and certain brain regions. Moreover, these results provided knowledge into how metastases can interrupt neurological functions, particularly in regions involved in coordination. Furthermore, combining neuroimaging with DEG and ROI-based feature selection revealed imaging features that are both clinically meaningful and biologically interpretable. Moreover, the study demonstrated that statistically grounded machine learning approaches can effectively address the challenges of high-dimensional radiogenomic data. By integrating regularisation, nonlinear modelling, and biologically informed feature selection, the proposed approach achieved a balance between discriminative performance and interpretability. The study therefore shows the potential of radiogenomics not only for classification but also for providing information into the ways through which metastases influence brain function and specific regions.

6.1.1 Limitations

The study's major limitation was UK Biobank which only had 17 genes which could have led to the loss of the most important genes. Secondly, the genomic analysis was based on small public datasets (16 paired samples in GSE125989), which could have limited the ability to detect meaningful expression changes. The relatively small sample size increased the risk of overfitting and limited the stability of model estimates. The integration of separate datasets also introduced potential bias, and the use of internal validation limited generalisability. In addition, feature importance measures remain associational and do not support causal inference. Lastly, modality- or atlas-specific DEG–NIDP associations were only seen when combining all imaging modalities and brain atlases rather than analyzing individual modalities or atlases separately. The small number of cases within each modality made it difficult to detect modality-specific associations.

6.1.2 Future Directions

Future work should extend the current modelling approach by incorporating all brain regions linked to the identified genes to assess whether this improves predictive performance and interpretability. The use of larger and more heterogeneous datasets is also necessary to improve statistical power and reduce the risk of overfitting. Moreover, the integration of additional genomic variables such as mutation profiles and measures of gene activity may provide a more comprehensive representation of tumor biology. Additionally, future research should extend this approach by exploring alternative regularisation methods such as Elastic Net, which may better handle correlated predictors. Bayesian approaches could also provide an approach for uncertainty quantification, while causal inference methods may offer deeper knowledge into gene -imaging correlations. External validation on independent datasets is essential to assess generalisability to enhance the robustness of radiogenomic models. Such approaches could bridge the interpretability gap that currently limits the adoption of AI models in the field of cancer research. Finally, this study shows that radiogenomic approaches, when applied to datasets such as the TCIA BCBM-RadioGenomics used in this work, can potentially bridge the gap between classification and explanation. This work has laid a strong foundation for future statistical tools that may aid in early detection, personalized monitoring, and improved care for patients at risk of BCBM.

7. R code

The complete R analysis pipeline used in this study is available at: GitHub Repository.

Bibliography

Achal Singh Achrol, Robert C Rennert, Carey Anders, Riccardo Soffietti, Manmeet S Ahluwalia, Lakshmi Nayak, Solange Peters, Nils D Arvold, Griffith R Harsh, Patricia S Steeg, et al. Brain metastases. *Nature Reviews Disease Primers*, page 5, 2019.

Hugo JWL Aerts, Emmanuel Rios Velazquez, Ralph TH Leijenaar, Chintan Parmar, Patrick Grossmann, Sara Carvalho, Johan Bussink, René Monshouwer, Benjamin Haibe-Kains, Derek Rietveld, et al. Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach. *Nature communications*, 2014.

S. Agarwal et al. A review of performance metrics for classification models. *International Journal of Data Science*, 2021.

Andre Altmann, Livia Tolosi, Oliver Sander, and Thomas Lengauer. Permutation importance: A corrected feature importance measure. *Bioinformatics*, 2010.

Francesca Angelone, Alfonso Maria Ponsiglione, Carlo Ricciardi, Maria Paola Belfiore, Gianluca Gatta, Roberto Grassi, Francesco Amato, and Mario Sansone. A machine learning approach for breast cancer risk prediction in digital mammography†. *Applied Sciences*, 2024.

Dooman Arefan. Machine learning prediction of axillary lymph node metastasis in breast cancer: 2d versus 3d radiomic features. *Medical physics*, 2020.

Michael A Babyak. What you see may not be what you get: a brief, nontechnical introduction to overfitting in regression-type models. *Biopsychosocial Science and Medicine*, 66(3):411–421, 2004.

Lucía Barbadilla-Martínez, Noud Klaassen, Bas van Steensel, and Jeroen de Ridder. Predicting gene expression from dna sequence using deep learning models. *Nature Reviews Genetics*, 2025.

Xavier Bledsoe and Eric R Gamazon. Neuroimagene: an r package for assessing the neurological correlates of genetically regulated gene expression. *BMC bioinformatics*, 2024.

Alice Boire, Priscilla K. Brastianos, Livia Garzia, and Manuel Valiente. Brain metastasis. *Nature Reviews Cancer*, 2020.

Peter D Bos, Xiaochun H-F Zhang, Cristina Nadal, and et al. Genes that mediate breast cancer metastasis to the brain. *Nature*, 2009.

Leo Breiman. Random forests. *Machine learning*, 2001.

Clare Bycroft, Colin Freeman, Desislava Petkova, Gavin Band, Lloyd T Elliott, Kevin Sharp, Allan Motyer, Damjan Vukcevic, Olivier Delaneau, Jared O'Connell, et al. The uk biobank resource with deep phenotyping and genomic data. *Nature*, 2018.

L. Cama et al. Explainable artificial intelligence in healthcare: A review. *Artificial Intelligence in Medicine*, 2025.

Diogo V Carvalho, Eduardo M Pereira, and Jaime S Cardoso. Machine learning interpretability: A survey on methods and metrics. *Electronics*, 8(8):832, 2019.

Eduarda Carvalho, Sule Canberk, Fernando Schmitt, and Nuno Vale. Molecular subtypes and mechanisms of breast cancer: Precision medicine approaches for targeted therapies. *Cancers*, 2025.

Soonmee Cha. Neuroimaging in neuro-oncology. *Neurotherapeutics*, 2009.

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 2016.

Xiaoyun Cheng and Mien-Chie Hung. Breast cancer brain metastases. *Cancer and Metastasis Reviews*, 2007.

Davide Chicco and Giuseppe Jurman. The matthews correlation coefficient (mcc) is more informative than accuracy and f1 score in binary classification evaluation. *BMC Genomics*, 2021.

Emily Clough and Tanya Barrett. The gene expression omnibus database. *Statistical Genomics: Methods and Protocols*, 2016.

Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine learning*, 1995.

Christina Curtis. Genomic profiling of breast cancers. *Current Opinion in Obstetrics and Gynecology*, 2015.

Pierre Lafaye de Micheaux, Rémy Drouilhet, and Benoit Liquet. *The R software: fundamentals of programming and statistical analysis*. Springer Science & Business, 2014.

Demetra Demetriou, Zarina Lockhat, Luke Brzozowski, Kamal S Saini, Zodwa Dlamini, and Rodney Hull. The convergence of radiology and genomics: Advancing breast cancer diagnosis with radiogenomics. *Cancers*, 2024.

Glynn Dennis Jr, Brad T Sherman, Douglas A Hosack, Jun Yang, Wei Gao, H Clifford Lane, and Richard A Lempicki. David: database for annotation, visualization, and integrated discovery. *Genome biology*, 2003.

Akshat Dubey, Suyash Agrawal, Varun Agrawal, Tanishq Dubey, and Arpita Jaiswal. Breast cancer and the brain: A comprehensive review of neurological complications. *Cureus*, 2023.

Ron Edgar, Michael Domrachev, and Alex E Lash. Gene expression omnibus: Ncbi gene expression and hybridization array data repository. *Nucleic acids research*, 2002a.

Ron Edgar, Michael Domrachev, and Alex E. Lash. Gene expression omnibus: Ncbi gene expression and hybridization array data repository. https://www.ncbi.nlm.nih.gov/geo/, 2002b.

Bradley J Erickson, Petros Korfiatis, Zeynettin Akkus, and Timothy L Kline. Machine learning for medical imaging. *Radiographics*, 2017.

Ming Fan, Pingping Xia, Robert Clarke, Yue Wang, and Lihua Li. Radiogenomic signatures reveal multiscale intratumour heterogeneity associated with biological functions and survival in breast cancer. *Nature communications*, 2020.

Tom Fawcett. An introduction to roc analysis. *Pattern Recognition Letters*, 2006.

Jerome H Friedman, Trevor Hastie, and Rob Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of statistical software*, 2010.

D. Fryer, I. Strümke, and H. Nguyen. Shapley values for feature selection: The good, the bad, and the axioms. *IEEE Access*, 2021. URL https://ieeexplore.ieee.org/document/9565902.

Bertoli Gloria Porro Danilo Gallivanone, Francesca. Radiogenomics, breast cancer diagnosis and characterization: current status and future directions. *Methods and Protocols*, 2022.

Ornella Garrone, Fiorella Ruatta, Carmen Giusy Rea, Nerina Denaro, Michele Ghidini, Carolina Cauchi, Claudia Bareggi, Barbara Galassi, Marco C Merlano, and Roberto Rosenfeld. Current evidence in the systemic treatment of brain metastases from breast cancer and future perspectives on new drugs, combinations and administration routes: A narrative review. *Cancers*, 2024.

Marco Gerlinger, Andrew J Rowan, Stuart Horswell, James Larkin, David Endesfelder, Eva Gronroos, Pierre Martinez, Nicholas Matthews, Aengus Stewart, Patrick Tarpey, et al. Intratumor heterogeneity and branched evolution revealed by multiregion sequencing. *New England journal of medicine*, 2012.

A Giannoudis, ES Sokol, T Bhogal, SH Ramkissoon, ED Razis, R Bartsch, JA Shaw, K McGregor, Alison Clark, RSP Huang, et al. Breast cancer brain metastases genomic profiling identifies alterations targetable by immune-checkpoint and parp inhibitors. *NPJ precision oncology*, 2024.

Robert J Gillies, Paul E Kinahan, and Hedvig Hricak. Radiomics: Images are more than pictures, they are data. *Radiology*, 2016.

Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016. URL https://www.deeplearningbook.org/.

Isabelle Guyon, Jason Weston, Stephen Barnhill, and Vladimir Vapnik. Gene selection for cancer classification using support vector machines. *Machine learning*, 2002.

James A. Hanley and Barbara J. McNeil. The meaning and use of the area under a receiver operating characteristic (roc) curve. *Radiology*, 1982.

Robert M Haralick, Karthikeyan Shanmugam, and Its' Hak Dinstein. Textural features for image classification. *IEEE Transactions on systems, man, and cybernetics*, 2007.

Frank E Harrell et al. *Regression modeling strategies: with applications to linear models, logistic regression, and survival analysis*. Springer, 2001.

Hayat. Cancer statistics, trends, and multiple primary cancer analyses from the surveillance, epidemiology, and end results (seer) program. *The oncologist*, 2007.

Wenle He, Wenhui Huang, Lu Zhang, Xuewei Wu, Shuixing Zhang, and Bin Zhang. Radiogenomics: bridging the gap between imaging and genomics for precision oncology. *MedComm*, 2024.

David W Hosmer Jr, Stanley Lemeshow, and Rodney X Sturdivant. *Applied logistic regression*. John Wiley & Sons, 2013.

Da Wei Huang, Brad T Sherman, and Richard A Lempicki. Systematic and integrative analysis of large gene lists using david bioinformatics resources. *Nature protocols*, 2009.

Mariarosaria Incoronato, Marco Aiello, Teresa Infante, Carlo Cavaliere, Anna Maria Grimaldi, Peppino Mirabelli, Serena Monti, and Marco Salvatore. Radiogenomic analysis of oncological data: a technical survey. *International journal of molecular sciences*, 2017.

Gabriel Kallah-Dagadu, Mohanad Mohammed, Justine B Nasejje, Nobuhle Nokubonga Mchunu, Halima S Twabi, Jesca Mercy Batidzirai, Geoffrey Chiyuzga Singini, Portia Nevhungoni, and Innocent Maposa. Breast cancer prediction based on gene expression data using interpretable machine learning techniques. *Scientific Reports*, 2025.

Christoph A Karlo, Pier Luigi Di Paolo, Joshua Chaim, A Ari Hakimi, Irina Ostrovnaya, Paul Russo, Hedvig Hricak, Robert Motzer, James J Hsieh, and Oguz Akin. Radiogenomics of clear cell renal cell carcinoma: associations between ct imaging features and mutations. *Radiology*, 2014.

G Khyathi, KP Indumathi, A Jumana Hasin, M Lisa Flavin Jency, G Krishnaprakash, FLAVIN JENCY M LISA, et al. Support vector machines: a literature review on their application in analyzing mass data for public health. *Cureus*, 2025.

Michael D Kuo and Neema Jamshidi. Behind the numbers: decoding molecular phenotypes with radiogenomics—guiding principles and technical considerations. *Radiology*, 2014.

Philippe Lambin, Emmanuel Rios-Velazquez, Ralph Leijenaar, Sara Carvalho, Ruud GPM Van Stiphout, Patrick Granton, Catharina ML Zegers, Robert Gillies, Ronald Boellard, André Dekker, et al. Radiomics: extracting more information from medical images using advanced feature analysis. *European journal of cancer*, 2012.

José Pablo Leone, Adrian V Lee, and Adam M Brufsky. Prognostic factors and survival of patients with brain metastasis from breast cancer who underwent craniotomy. *Cancer medicine*, 2015.

Li. A review of radiomics and genomics applications in cancers: the way towards precision medicine. *Radiation Oncology*, 2022.

Chaofan Li, Mengjie Liu, Yinbin Zhang, Yusheng Wang, Jia Li, Shiyu Sun, Xuanyu Liu, Huizi Wu, Cong Feng, Peizhuo Yao, et al. Novel models by machine learning to predict prognosis of breast cancer brain metastases. *Journal of translational medicine*, 2023.

Nancy U Lin and Eric P Winer. Brain metastases: the her2 paradigm. *Clinical Cancer Research*, 2007.

Roberto Lo Gullo, Isaac Daimiel, Elizabeth A Morris, and Katja Pinker. Combining molecular and imaging metrics in cancer: radiogenomics. *Insights into imaging*, 2020.

Wei-Yin Loh. Classification and regression trees. *Wiley interdisciplinary reviews: data mining and knowledge discovery*, 2011.

Scott Lundberg, Gabriel Erion, and Su-In Lee. Consistent individualized feature attribution for tree ensembles. 2018. doi: 10.48550/arXiv.1802.03888.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 2017.

Xiao Luo, Hui Xie, Yadi Yang, Cheng Zhang, Yijun Zhang, Yue Li, Qiuxia Yang, Deling Wang, Yingwei Luo, Zhijun Mai, et al. Radiomic signatures for predicting receptor status in breast cancer brain metastases. *Frontiers in Oncology*, 2022.

Maciej A Mazurowski. Radiogenomics: what it is and why it is important. *Journal of the American College of Radiology*, 2015.

Valerie McCormack, Fiona McKenzie, Milena Foerster, Annelle Zietsman, Moses Galukande, Charles Adisa, Angelica Anele, Groesbeck Parham, Leeya F Pinder, Herbert Cubasch, et al. Breast cancer survival and survival gap apportionment in sub-saharan africa (abc-do): a prospective cohort study. *The Lancet Global Health*, 2020.

Johns Hopkins Medicine. Brain anatomy and how the brain works, 2025. URL https://www.hopkinsmedicine.org/health/conditions-and-diseases/anatomy-of-the-brain.

David Meyer, Evgenia Dimitriadou, Kurt Hornik, Andreas Weingessel, Friedrich Leisch, Chih-Chung Chang, and Chih-Jen Lin. *e1071: Misc Functions of the Department of Statistics, Probability Theory Group (Formerly: E1071), TU Wien*, 2019. URL https://CRAN.R-project.org/package=e1071. R package version 1.7-3.

Monica Morrow, Janet Waters, and Elizabeth Morris. Mri for breast cancer screening, diagnosis, and treatment. *The Lancet*, 2011.

K. Naidu et al. Evaluation metrics for machine learning models in medical diagnosis. *Journal of Healthcare Informatics Research*, 2023.

National Human Genome Research Institute. Gene - national human genome research institute, 2023. URL https://www.genome.gov/genetics-glossary/Gene. Accessed: 2025-04-24.

Lakshmi Nayak, Eudocia Quant Lee, and Patrick Y Wen. Epidemiology of brain metastases. *Current oncology reports*, 2012.

John Ashworth Nelder and Robert WM Wedderburn. Generalized linear models. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 1972.

Anna Niwińska, Magdalena Murawska, and Katarzyna Pogoda. Breast cancer brain metastases: differences in survival depending on biological subtype, rpa rtog prognostic class and systemic treatment after whole-brain radiotherapy (wbrt). *Annals of Oncology*, 2010.

William Stafford Noble et al. Support vector machine applications in computational biology. *Kernel methods in computational biology*, 2004.

M. Olson and et al. Her2-targeted therapies in breast cancer: A review of current and future strategies. *Journal of Clinical Oncology*, 2013.

Erasmo Orrantia-Borunda, Patricia Anchondo-Nuñez, Lucero Evelia Acuña-Aguilar, Francisco Octavio Gómez-Valles, and Claudia Adriana Ramírez-Valdespino. Subtypes of breast cancer. *Breast Cancer [Internet]*, 2022.

Matthew R Orton, Evan Hann, Simon J Doran, Scott TC Shepherd, Derfel Ap Dafydd, Charlotte E Spencer, José I López, Víctor Albarrán-Artahona, Francesca Comito, Hannah Warren, et al. Interpretability of radiomics models is improved when using feature group selection strategies for predicting molecular and clinical targets in clear-cell renal cell carcinoma: Insights from the tracerx renal study. *Cancer Imaging*, 2023.

Ah Young Park, Mi-Ryung Han, Bo Kyoung Seo, Hye-Yeon Ju, Gil Soo Son, Hye Yoon Lee, Young Woo Chang, Jungyoon Choi, Kyu Ran Cho, Sung Eun Song, et al. Mri-based breast cancer radiogenomics using rna profiling: association with subtypes in a single-center prospective study. *Breast Cancer Research*, 2023.

Eun Kyung Park, Kwang-sig Lee, Bo Kyoung Seo, Kyu Ran Cho, Ok Hee Woo, Gil Soo Son, Hye Yoon Lee, and Young Woo Chang. Machine learning approaches to radiogenomics of breast cancer using low-dose perfusion computed tomography: predicting prognostic biomarkers and molecular subtypes. *Scientific reports*, 2019.

Chintan Parmar, Patrick Grossmann, Derek Rietveld, Michelle M Rietbergen, Philippe Lambin, and Hugo JWL Aerts. Radiomic machine-learning classifiers for prognostic biomarkers of head and neck cancer. *Frontiers in oncology*, 2015.

Helen Pearson. Genetics: what is a gene? *Nature*, 2006.

Charles M Perou, Therese Sørlie, Michael B Eisen, Matt Van De Rijn, Stefanie S Jeffrey, Christian A Rees, Jonathan R Pollack, Douglas T Ross, Hilde Johnsen, Lars A Akslen, et al. Molecular portraits of human breast tumours. *nature*, 2000.

Matthew E Ritchie, Belinda Phipson, DI Wu, Yifang Hu, Charity W Law, Wei Shi, and Gordon K Smyth. limma powers differential expression analyses for rna-sequencing and microarray studies. *Nucleic acids research*, 2015.

Maria Rossing, Claus Storgaard Sørensen, Bent Ejlertsen, and Finn Cilius Nielsen. Whole genome sequencing of breast cancer. *Apmis*, 2019.

Rezvan Rostami, Shivam Mittal, Pooya Rostami, Fattaneh Tavassoli, and Bahman Jabbari. Brain metastasis in breast cancer: a comprehensive literature review. *Journal of neuro-oncology*, 2016.

Benedek Rozemberczki, Lauren Watson, Péter Bayer, Hao-Tsung Yang, Olivér Kiss, Sebastian Nilsson, and Rik Sarkar. The shapley value in machine learning. In *The 31st International Joint Conference on Artificial Intelligence and the 25th European Conference on Artificial Intelligence*. International Joint Conferences on Artificial Intelligence Organization, 2022.

David Ruppert. The elements of statistical learning: data mining, inference, and prediction, 2004.

Ashirbani Saha, Michael R Harowicz, Lars J Grimm, Connie E Kim, Sujata V Ghate, Ruth Walsh, and Maciej A Mazurowski. A machine learning approach to radiogenomics of breast cancer: a study of 922 subjects and 529 dce-mri features. *British journal of cancer*, 2018.

Takaya Saito and Marc Rehmsmeier. The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 2015.

Isabella Sanchez and Ruman Rahman. Radiogenomics as an integrated approach to glioblastoma precision medicine. *Current Oncology Reports*, 2024.

S Sathyanarayanan and B Roopashri Tantri. Confusion matrix-based performance evaluation metrics. *African Journal of Biomedical Research*, 2024.

Bernhard Schölkopf and Alexander J Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2002.

C. Severn, K. Suresh, C. Görg, Y. S. Choi, R. Jain, and D. Ghosh. A pipeline for the implementation and visualization of explainable machine learning for medical imaging using radiomics features. *Sensors*, 2022.

A. Singh, S. Sengupta, and V. Lakshminarayanan. Explainable deep learning models in medical image analysis. *Journal of Imaging*, 2020.

Douglas Sloane and S Philip Morgan. An introduction to categorical data analysis. *Annual review of sociology*, 1996.

Gordon K Smyth, Natalie Thorne, and James Wettenhall. Limma: linear models for microarray data user's guide. *Software manual available from http://www. bioconductor. org*, 2003.

Clemens P Spielvogel, Stefan Stoiber, Laszlo Papp, Denis Krajnc, Marko Grahovac, Elisabeth Gurnhofer, Karolina Trachtova, Vojtech Bystry, Asha Leisser, Bernhard Jank, et al. Radiogenomic markers enable risk stratification and inference of mutational pathway states in head and neck cancer. *European Journal of Nuclear Medicine and Molecular Imaging*, 2023.

B. Taha, D. Wu, L. Sabal, M. Kollitz, A. Venteicher, and Y. Watanabe. Mri dataset of metastatic breast cancer to the brain with expert-reviewed segmentations and tumor-derived radiomic features (bcbm-radiogenomics) (version 1), 2025.

M Teja, Y Yamini Priyanka, and N Mehar Jyothi. Enhanced drowsiness detection accident prevention system via facial features 1. Technical report, 2024. URL www.iosrjen.org.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 1996.

Vladimir N Vapnik. A note on one class of perceptrons. *Automat. Rem. Control*, 1964.

Shuo Wang, Meng Liu, Yan Li, Xinyu Zhang, Mengting Sun, Zian Wang, Ruokun Li, Qirong Li, Qing Li, Yili He, et al. Recent advances in data-driven fusion of multi-modal imaging and genomics for precision medicine. *Information Fusion*, 2025.

J Yan, B Zhang, S Zhang, J Cheng, X Liu, W Wang, Y Dong, L Zhang, X Mo, Q Chen, et al. Quantitative mri-based radiomics for noninvasively predicting molecular subtypes and survival in glioma patients. npj precis oncol. 2021; 5: 72. *J Neuroimmunol*, 2011.

Albert C Yeh, Hui Li, Yitan Zhu, Jing Zhang, Galina Khramtsova, Karen Drukker, Alexandra Edwards, Stephanie McGregor, Toshio Yoshimatsu, Yonglan Zheng, et al. Radiogenomics of breast cancer using dynamic contrast enhanced mri and gene expression profiling. *Cancer Imaging*, 2019.

SM Yu, YH Chan, YS Chan, C Tsoi, GKF Tam, EHY Hung, WCW Chu, and HHL Chau. Molecular classification and respective radiological phenotypes of breast cancers: a pictorial essay. *Hong Kong J. Radiol*, 2023.

Tiantian Zeng, Md Selim, Jie Zhang, Arnold Stromberg, Jin Chen, and Chi Wang. Multivariate sparse group lasso joint model for radiogenomics data. *arXiv preprint arXiv:2206.01352*, 2022.

Yumin Zhang, Haotian Shang, Jiaxuan Zhang, Yizhi Jiang, Jiahao Li, Huihua Xiong, and Tengfei Chao. Drug treatment direction based on the molecular mechanism of breast cancer brain metastasis. *Pharmaceuticals*, 2025. doi: 10.3390/ph18020262. URL https://www.mdpi.com/1424-8247/18/2/262.

Zhe Zhu, Ehab Albadawy, Ashirbani Saha, Jun Zhang, Michael R Harowicz, and Maciej A Mazurowski. Deep learning for identifying radiogenomic associations in breast cancer. *Computers in biology and medicine*, 2019.

Pascal O Zinn, Bhanu Majadan, Pratheesh Sathyan, Sanjay K Singh, Sadhan Majumder, Ferenc A Jolesz, and Rivka R Colen. Radiogenomic mapping of edema/-cellular invasion mri-phenotypes in glioblastoma multiforme. *PloS one*, 2011.

A. Zwanenburg and et al. The image biomarker standardisation initiative: Standardizing radiomic features for reproducibility. *Radiology*, 2020.

Alex Zwanenburg, Stefan Leger, Martin Vallières, and Steffen Löck. The image biomarker standardization initiative: Standardized quantitative radiomics for high-throughput image-based phenotyping. *Radiology*, 2020a.

Alex Zwanenburg, Martin Vallières, Mahmoud A Abdalah, Hugo JWL Aerts, Vincent Andrearczyk, Aditya Apte, Saeed Ashrafinia, Spyridon Bakas, Roelof J

Beukinga, Ronald Boellaard, et al. The image biomarker standardization initiative: standardized quantitative radiomics for high-throughput image-based phenotyping. *Radiology*, 2020b.